Unit

24 Bond polarity and intermolecular forces

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

- Electronegativity and bond polarity
- Van der Waals' forces
- Factors affecting the strength of van der Waals' forces
- Hydrogen bonding
- Surface tension and viscosity of liquids

24.1 The position of electrons in a covalent bond

In a hydrogen molecule, one pair of electrons is shared between the two hydrogen atoms. The extent of attraction from each hydrogen nucleus is the same because both nuclei are the same. The pair of electrons in the covalent bond is shared equally between two hydrogen atoms. We say that the covalent bond is **non-polar**.

H : H

the bonding electrons are shared equally between two H atoms

However, when the two atoms at each end of a covalent bond are different, the attractive forces by the nuclei may be different and the pair of electrons is no longer equally shared between the two atoms.

For example, in a hydrogen chloride molecule, the chlorine nucleus exerts a stronger attraction to the electrons in the covalent bond than the hydrogen nucleus does.

H :Cl

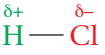
the bonding electrons are attracted more strongly by Cl than by H

Hence, the bonding electrons are closer to the chorine atom than to the hydrogen atom. We say that the covalent bond is **polar**.

non-polar 非極性　　polar 極性

✓ Polar covalent bonding occurs because one bonded atom has a stronger attraction for electrons than the other bonded atom does.

In a hydrogen chloride molecule, the chlorine atom gets a slightly negative charge because it has a greater share of the bonding electrons. The hydrogen atom becomes slightly positively charged because it has lost some of its share in the bonding electrons. We can show the small amount of electrical charge by δ– and δ+ where δ means 'a small amount of'.



24.2 Electronegativity

In a covalent bond between two different atoms, the sharing of electrons is likely to be unequal because the different atoms have different powers to attract the bonding electrons. The term used to describe such a power is **electronegativity**.

✓ The electronegativity of an element represents the power of an atom of that element to attract a bonding pair of electrons towards itself in a molecule.

Electronegativity and the periodic table

There are various scales used to measure electronegativity. The one most commonly used was devised by Linus Pauling, and is called the Pauling scale of electronegativity. It is a relative scale and so has no units. It runs from 0 to 4. The greater is the value of the electronegativity, the greater is the power of an atom to attract the electrons in a covalent bond towards itself.

electronegativity 電負性

The noble gases have no electronegativity values because they do not, in general, form covalent bonds.

Fig. 24.1 shows the electronegativities of some common elements in the periodic table. They are based on the Pauling scale.

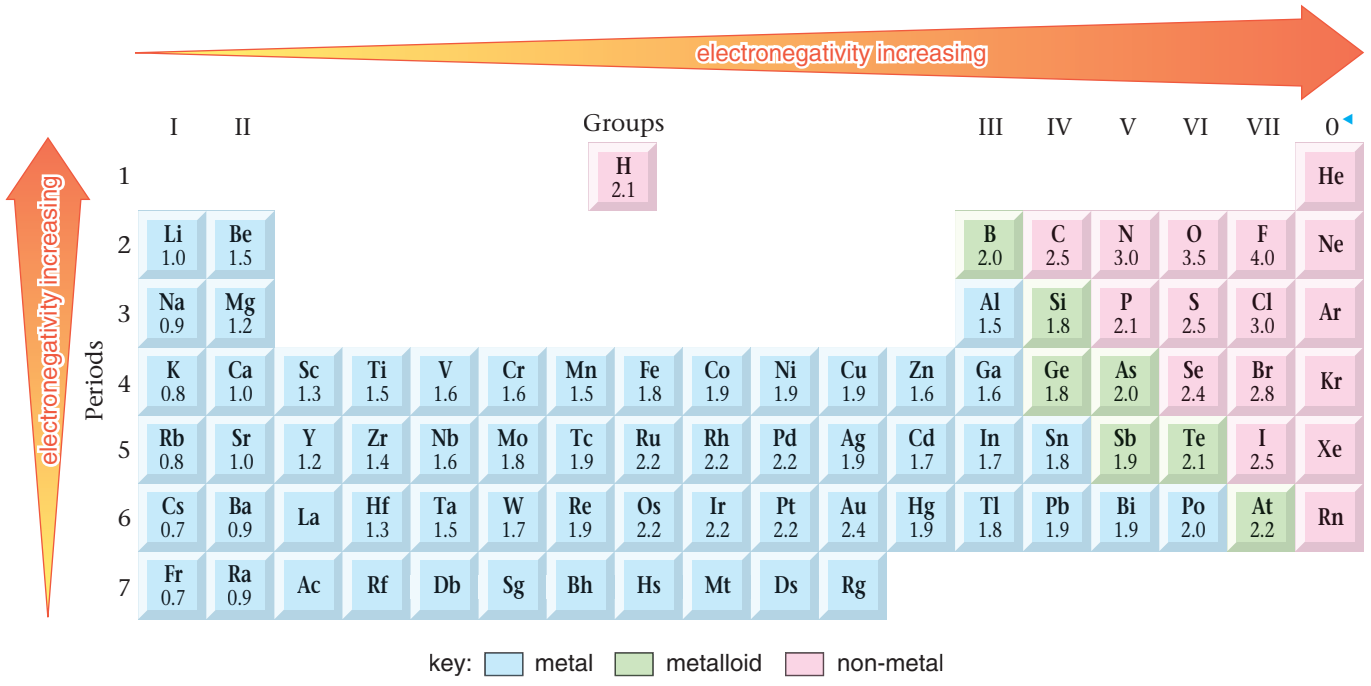


Fig. 24.1 The electronegativities of common elements on the Pauling scale

The trends do not apply to transition metals and noble gases.

A careful examination of Fig. 24.1 reveals trends and relationships among the electronegativities of different elements.

- Non-metals have higher electronegativities than metals.
- Electronegativity increases from left to right across a period of the periodic table.
- Within each group of the main group elements, electronegativity decreases with increasing atomic number.
- The most electronegative element is fluorine.

N1 Refer to page T17.

We can use the difference in electronegativity between two bonded atoms to predict how polar the covalent bond between them will be. For example, the electronegativity of carbon is 2.5 and that of fluorine is 4.0. There is a large difference for electronegativity between carbon and fluorine. So the fluorine atom attracts the bonding electrons more strongly than the carbon atom does and the C–F bond is polar. N1

$$\overset{\delta+}{C}\text{—}\overset{\delta-}{F}$$

On the other hand, a C–S bond is not polar because carbon and sulphur have the same electronegativity.

Dipole and dipole moment

A pair of equal and opposite charges separated by a distance as in the H–Cl molecule is call a **dipole**. The hydrogen chloride molecule discussed in section 24.1 has a dipole.

A quantitative measure of the polarity of a bond is its **dipole moment** — the magnitude of either of the charges multiplied by the distance between the centres of the charges.

We can represent the direction of the dipole moment by an arrow ($\dashrightarrow$) pointing from the slightly positive end to the slightly negative end. For example,

$$\overset{\delta+}{H}\text{—}\overset{\delta-}{Cl}$$
$$\dashrightarrow$$

Practice 24.1

Decide whether each of the following bonds is polar or non-polar. Where appropriate, show the slightly negatively charged end using δ– and the slightly positively charged end using δ+.

a) P–H
non-polar

b) $\overset{\delta+}{H}\text{–}\overset{\delta-}{F}$

c) $\overset{\delta-}{O}\text{–}\overset{\delta+}{N}$

d) F–F
non-polar

e) $\overset{\delta-}{N}\text{–}\overset{\delta+}{H}$

f) C–I
non-polar

dipole 偶極　　dipole moment 偶極矩



Fig. 24.2 A stream of liquid deflected by a negatively charged rod

24.3 Effect of a charged rod on polar and non-polar liquids

When two different atoms with different electronegativities are joined by a covalent bond, the more electronegative atom exerts a stronger attraction to the bonding electrons and a polar bond results.

It is possible to have several polar bonds in a molecule. Does this make the whole molecule polar? We can perform the following experiment to test whether or not molecules of a liquid are polar:

1 Rub a polythene rod with fur. This will give the rod a negative charge.

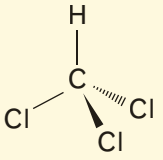
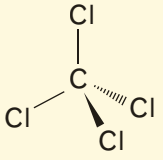
2 Bring the charged rod close to a stream of the liquid from a burette and notice if any deflection occurs (Fig. 24.2).

Table 24.1 shows the molecular formulae and 3-D structures of trichloromethane and tetrachloromethane.

24.1
Testing liquids to find out if their molecules are polar.

Video
Testing whether water molecules are polar

Table 24.1
The molecular formulae and 3-D structures of trichloromethane and tetrachloromethane

Compound	Molecular formula	3-D structure
Trichloromethane	$CHCl_3$	
Tetrachloromethane	CCl_4	

From the experiment, it is found that the stream of trichloromethane is deflected by the charged rod while that of tetrachloromethane is not.

The trichloromethane molecule

The electronegativity value of chlorine is greater than that of carbon. So each C–Cl bond in a trichloromethane molecule is polar.

The trichloromethane molecule is tetrahedral in shape. The individual C–Cl bond dipole moment reinforces each other. The molecule has a net dipole moment and it is polar (Fig. 24.3).

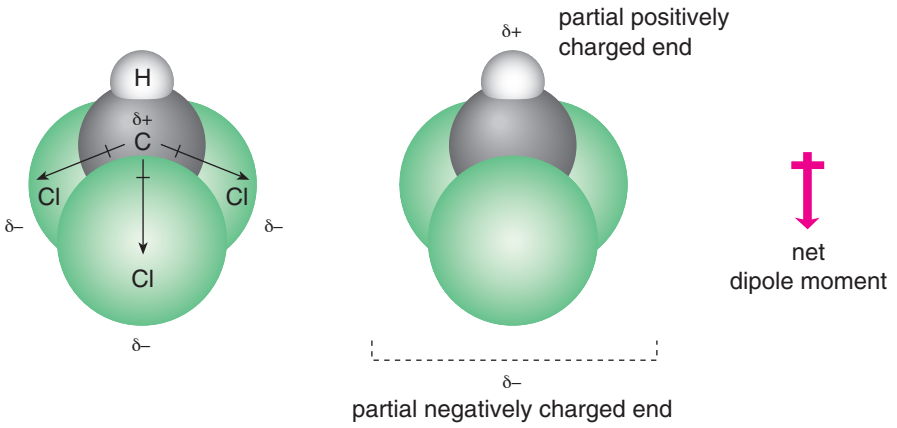


Fig. 24.3 The polar trichloromethane molecule

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Fig. 24.4 shows that a stream of trichloromethane is deflected by a negatively charged rod. When we bring the charged rod close to the liquid stream, positive ends of the molecules are attracted towards the rod.

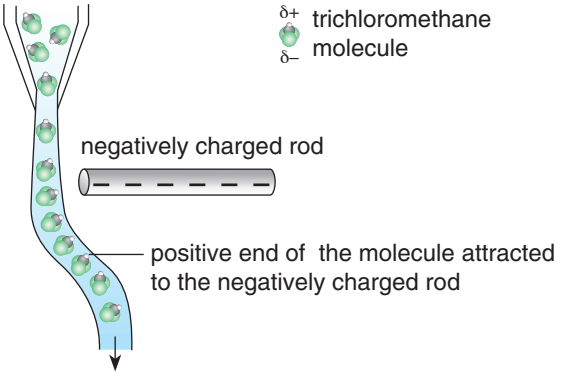


Fig. 24.4 Trichloromethane molecules are attracted by a negatively charged rod (This figure is not drawn to scale.)

If a positively charged rod were used in the experiment, the liquid stream would also be deflected towards the rod. However, here the negative ends of the trichloromethane molecules would be attracted towards the rod.

The tetrachloromethane molecule

A tetrachloromethane molecule has a tetrahedral shape. Due to the symmetry of the tetrahedral shape, the four identical C–Cl bond dipole moments cancel one another out exactly. As a result, the molecule has no net dipole moment and it is non-polar (Fig. 24.5). Thus, a stream of tetrachloromethane is not deflected by the charged rod. N2

N2 The polarity of the trichloromethane molecule and tetrachloromethane molecules may also be explained in the following way:

The bonds in both CH_3Cl and CCl_4 are polar. The CH_3Cl molecule is polar, because the polarity of the C–Cl bond is not cancelled out. However, the CCl_4 molecule is not polar because the polarities of the four C–Cl bonds cancel out since the molecule is symmetrical.

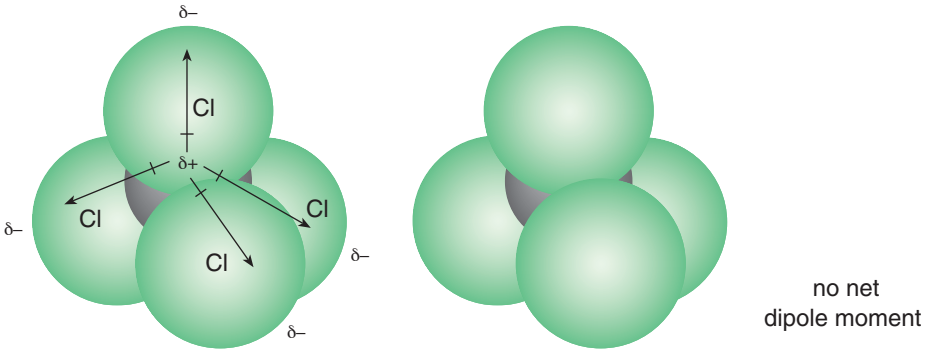


Fig. 24.5 The non-polar tetrachloromethane molecule

To summarize, for a molecule to be polar, the following conditions must apply:

- the molecule must contain one or more polar bonds;
- the shape of the molecule must not be completely symmetrical.

24.4 Common polar molecules

The water molecule

The electronegativity value of oxygen is greater than that of hydrogen. So each O–H bond in a water molecule is polar.

As the water molecule has a V-shape or bent shape, the individual bond dipole moments reinforce each other. The water molecule has a net dipole moment and it is polar (Fig. 24.6).

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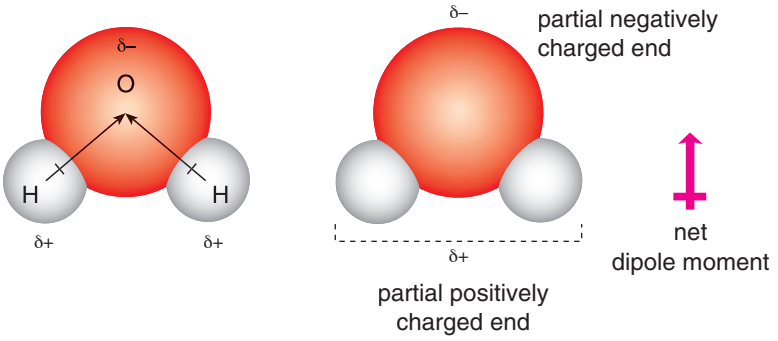


Fig. 24.6 The polar water molecule

In the water / ammonia molecule, every lone pair of electrons on the central atom would project out into space away from the positively charged nucleus. We can consider the nucleus 'end' of the lone pair as positive and the electron 'end' as negative. Such 'lone pair dipole' also contributes to the polarity of the molecule.

The ammonia molecule

The electronegativity value of nitrogen is greater than that of hydrogen. So each N–H bond in an ammonia molecule is polar.

As the ammonia molecule has a trigonal pyramidal shape, the individual bond dipole moments reinforce each other. The ammonia molecule has a net dipole moment and it is polar (Fig. 24.7).

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DSE 2013 Paper 1B Q2(b)

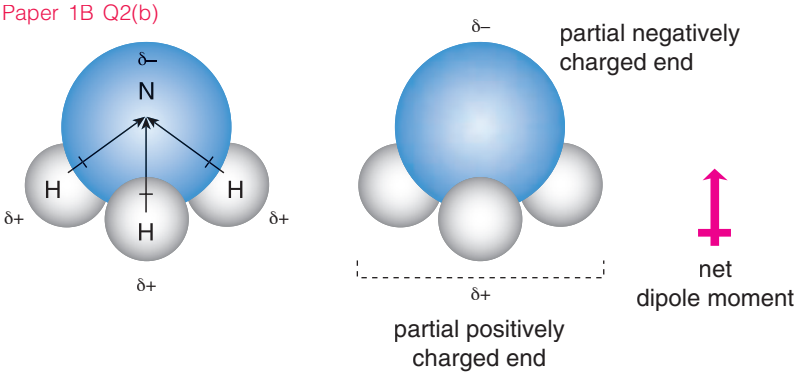


Fig. 24.7 The polar ammonia molecule

24.5 Common non-polar molecules

The carbon dioxide molecule

The electronegativity value of oxygen is greater than that of carbon. So each C=O bond in a carbon dioxide molecule is polar.

A carbon dioxide molecule is linear in shape. Fig. 24.8 shows that with the polar C=O bonds of the molecule arranged in a straight line, the two identical bond dipole moments cancel each other out exactly. As a result, the molecule has no net dipole moment and it is non-polar.

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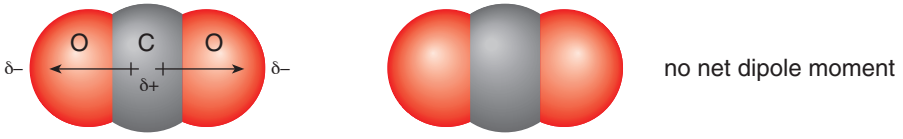


Fig. 24.8 The non-polar carbon dioxide molecule

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DSE 2013 Paper 1B Q2(b)
ASL 2008 Paper 1 Q5(b)

The boron trifluoride molecule

The electronegativity value of fluorine is greater than that of boron. So each B–F bond in a boron trifluoride molecule is polar.

N3 Examination questions often ask whether BF_3 molecule is polar.

A boron trifluoride molecule has a trigonal planar shape. Due to N3 the symmetry of a trigonal planar shape, the three identical bond dipole moments cancel one another out exactly (Fig. 24.9). As a result, the molecule has no net dipole moment and it is non-polar.

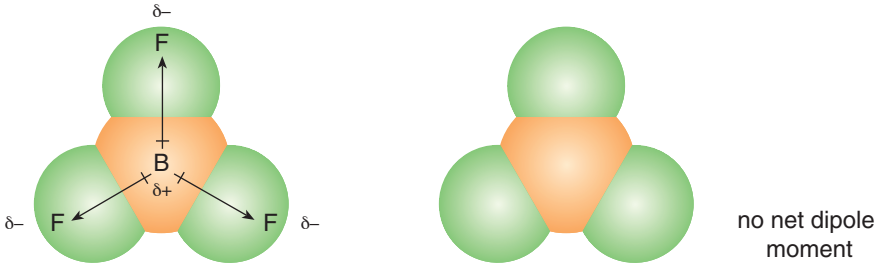


Fig. 24.9 The non-polar boron trifluoride molecule

Practice 24.2

Complete the table by filling in the following information for each molecule:

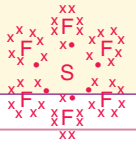
a) draw an electron diagram of the molecule;

b) write the name of the shape of the molecule;

c) give the polar bond in the molecule;

d) decide whether the molecule is polar.

Water is given as an example.

Molecular formula	Electron diagram	Shape of molecule	Polar bond	Polar molecule?
H_2O		V-shape or bent shape	O–H	yes
HF		linear	H–F	yes
CH_2F_2		tetrahedral	C–F	yes
SF_6		octahedral	S–F	no

24.6 Attractive forces between non-polar molecules

Noble gases such as neon and argon exist as separated atoms. Noble gases can be liquefied at very low temperatures. So there must be some attractive forces between their atoms. At a very low temperature, these attractive forces are strong enough to hold the atoms together, causing the gas to condense.

Bromine is a non-polar molecule. It is a liquid at room temperature. This suggests that there must be some attractive forces between bromine molecules. These **intermolecular forces** keep the molecules together as a liquid at room temperature.

intermolecular force 分子間引力

Consider a bromine molecule. Averaged over time, the distribution of electrons throughout the molecule is symmetrical. However, electrons are in constant motion. At a particular instant, there may be more electrons at one end of the molecule than at the other; instantaneously the molecule has developed a dipole, i.e. an **instantaneous dipole** (Fig. 24.10).

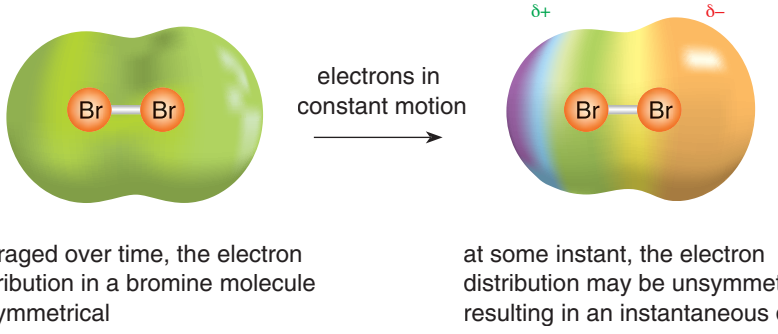


Fig. 24.10 How an instantaneous dipole forms

This figure employs a useful way of visualizing the uneven distribution of electrons in a molecule — using colour to portray the electron distribution. An orange colour is shown on the side with a slightly negative charge.

This instantaneous dipole can affect the electron distributions in neighbouring molecules and induce dipoles in these neighbours.

There are attractive forces between the δ+ end of the dipole in one molecule and the δ– end of the dipole in a neighbouring molecule (Fig. 24.11). Such attractions operate between all atoms or molecules.

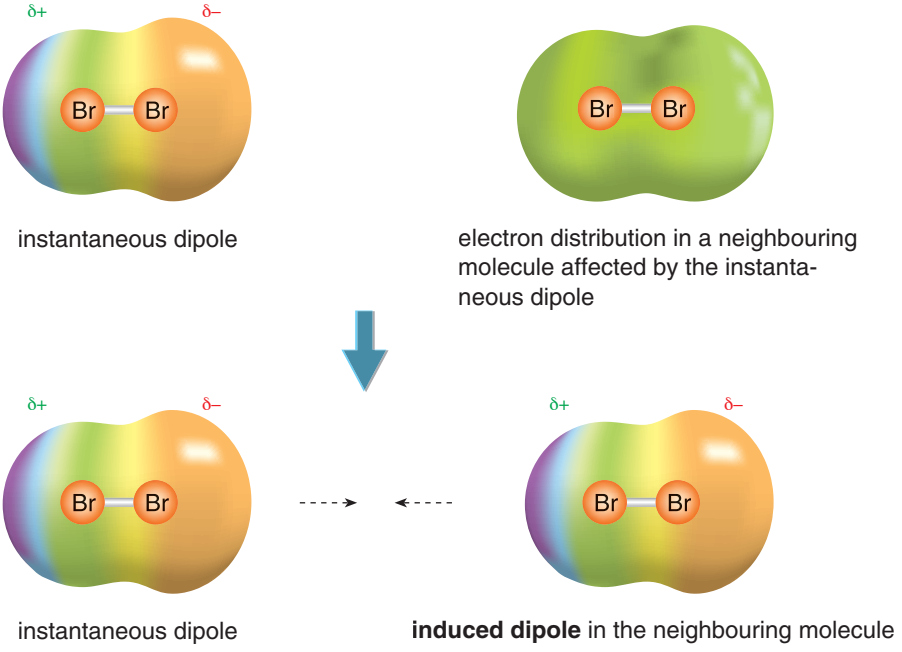


Fig. 24.11 The instantaneous dipole can induce a dipole in a neighbouring molecule

Reference Website
Refer to the following website for an animation showing the instantaneous dipole-induced dipole attractions between molecules:
http://antoine.frostburg.edu/chem/senese/101/liquids/faq/h-bonding-vs-london-forces.shtml
(accessed July 2014)

instantaneous dipole 瞬間偶極　　induced dipole 誘發偶極

24.7 Attractive forces between polar molecules N4

N4 Refer to page T18.

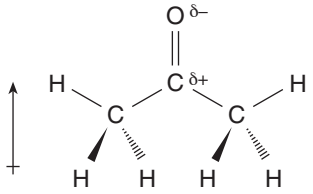


Fig. 24.12 The propanone molecule has a permanent dipole

As mentioned before, a molecule is polar if it contains polar bond(s) and the molecule is not completely symmetrical. Since uneven charge distribution exists all the time, the molecule as a whole is said to have a **permanent dipole**. For example, propanone molecules are polar and each of them has a permanent dipole (Fig. 24.12).

Propanone is a liquid at room temperature, and in the liquid state the molecules are constantly moving and tumbling around. Sometimes the negative end of one molecule will line up with the positive end of another, making them attract. Sometimes there will be two positive ends or two negative ends together, making the molecules repel (Fig. 24.13). Overall, there will be more attractions than repulsions between the permanent dipoles; that is the reason why the molecules stay together as a liquid.

Reference Website

Refer to the following websites for animations showing permanent dipole-permanent dipole attractions between molecules

- http://antoine.frostburg.edu/chem/senese/101/liquids/faq/h-bonding-vs-london-forces.shtml (accessed July 2014)
- http://chemmovies.unl.edu/ChemAnime/DIPOLED/DIPOLED.html (accessed July 2014)

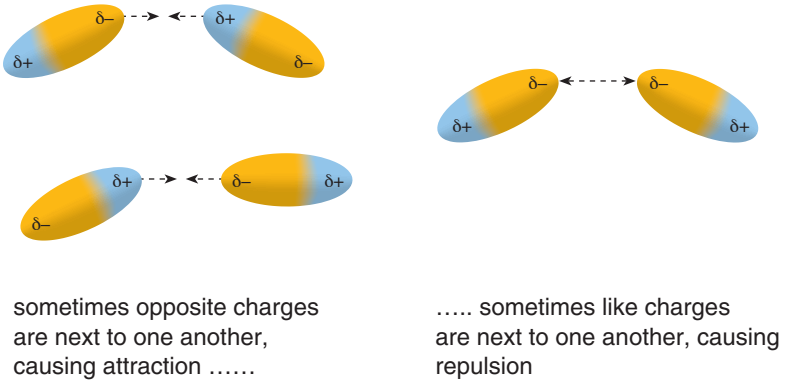


Fig. 24.13 Molecules in a liquid are constantly moving and tumbling around

Van der Waals' forces

The Dutch physicist J. D. van der Waals' postulated the existence of forces of attraction and repulsion that are neither ionic nor covalent. Such forces arise in a number of ways and are collectively called **van der Waals' forces**. Attractive forces between non-polar molecules and those between polar molecules are both van der Waals' forces.

permanent dipole 永久偶極　　van der Waals' forces 范德華力

24.8 Factors affecting the strength of van der Waals' forces

Number of electrons in the atom or molecule

A large molecule has many electrons. The outer electrons are further away from the nucleus, and thus held less tightly. The electron distribution in the molecule can be distorted easily. This favours the occurrence of instantaneous dipoles (and also favours the creation of induced dipoles). Thus, stronger van der Waals' forces exist between these molecules.

Thus, the strength of van der Waals' forces increases with the number of electrons present in the atom or molecule. This means that atoms or molecules with greater relative atomic or molecular masses produce stronger van der Waals' forces than atoms or molecules with small relative atomic or molecular masses. The strength of van der Waals' forces between molecules are also affected by molecular size for the same reason.

Boiling points of noble gases

Fig. 24.14 shows the boiling points of noble gases plotted against the numbers of electrons present. Notice that the boiling point increases as the number of electrons present increases. This is because the van der Waals' forces between the atoms of a noble gas increase with an increasing number of electrons present. So more heat is needed to separate the atoms of the noble gas with more electrons and the boiling point of the noble gas is higher.

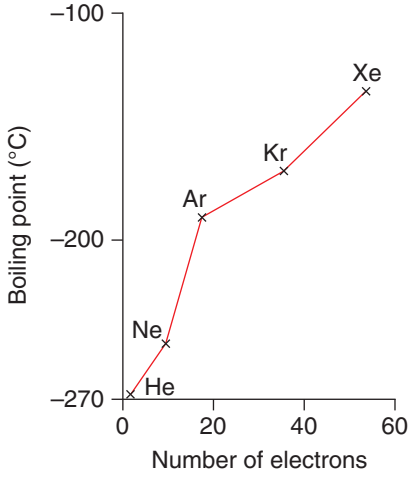


Fig. 24.14 Boiling points of noble gases plotted against the numbers of electrons present

Example 24.1

Q Some information about three hydrogen halides is shown below. Explain the trend in the boiling point of the hydrogen halides.

Hydrogen halide	Relative molecular mass	Boiling point (°C)
HCl	36.5	–85
HBr	80.9	–67
HI	127.9	–35

A The boiling point of a compound depends on the strength of its intermolecular attractions. The intermolecular attractions in these hydrogen halides are predominantly van der Waals' forces.

The number of electrons in one molecule increases from HCl to HI. Hence the strength of van der Waals' forces also increases from HCl to HI. This suggests that the boiling points should increase from HCl to HI, in accordance with the data.

Practice 24.3

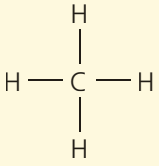
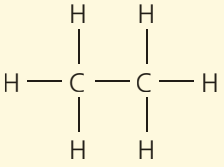
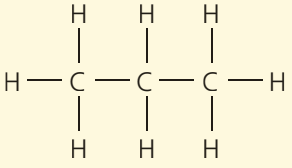
The boiling points of three simple carbon compounds are shown below. Explain the order of the boiling points in terms of intermolecular forces.

The boiling point of a carbon compound depends on the strength of its intermolecular attractions.

The intermolecular attractions in the carbon compounds are van der Waals' forces.

The number of electrons in the molecule / the molecular mass increases from methane to propane.

Hence the strength of van der Waals' forces also increases from methane to propane. This suggests that the boiling points should increase from methane to propane, in accordance with the data.

Compound	Structure	Boiling point (°C)
Methane		–162
Ethane		–89
Propane		–42

Shape of the molecule

The shape of the molecule is also important in determining the magnitude of van der Waals' forces between molecules.

We will further discuss pentane and 2,2-dimethylpropane in Topic 7 Fossil Fuels and Carbon Compounds.

Fig. 24.15 lists the different boiling points of pentane and 2,2-dimethylpropane (or dimethylpropane), which are 36 °C and 10 °C respectively. These compounds have the same molecular formula of C_5H_{12}. Thus, they have equal number of electrons in their molecules.

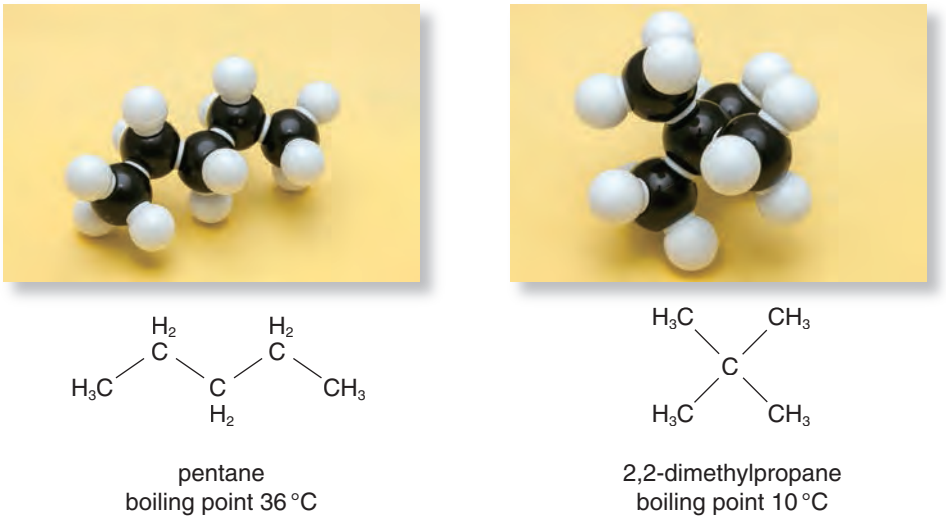


Fig. 24.15 Comparing the boiling points of pentane and 2,2-dimethylpropane

The pentane molecule is longer and somewhat spreadout. It has a greater molecular surface area and allows greater contact with neighbouring molecules (Fig. 24.16a). The 2,2-dimethylpropane molecule is more spherical and compact. The molecular surface area available for coming into contact with neighbouring molecules is smaller (Fig. 24.16b). The van der Waals' forces between pentane molecules are thus stronger. Hence pentane has a higher boiling point. N5

N5 Comparing the densities of pentane and 2,2-dimethylpropane

Compound	Density (g cm^{-3})
pentane	0.626
2,2-dimethylpropane	0.591

The stronger van der Waals' forces between pentane molecules pull the molecules closer together. So, the density of pentane is higher.

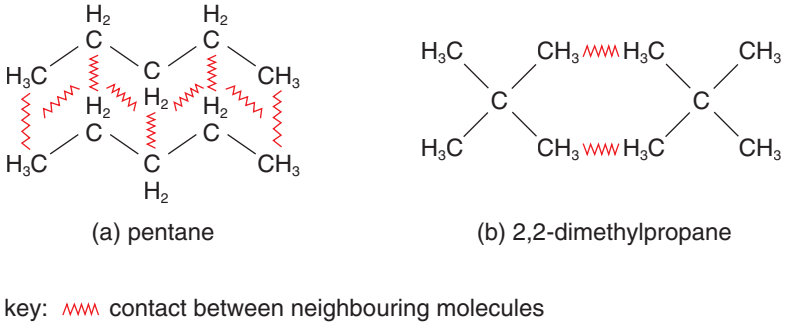


Fig. 24.16 Contact with neighbouring molecules in pentane and 2,2-dimethylpropane

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24.9 Hydrogen bonding

ASL 2008 Paper 1 Q10

How does hydrogen bonding affect the boiling point of hydrogen fluoride?

Fig. 24.17 shows the boiling points of the hydrogen halides, HF, HCl, HBr and HI. The rise in the boiling point from HCl to HI is due to the increasing number of electrons present in the halogen atoms as we go down the group. This leads to increasing van der Waals' forces.

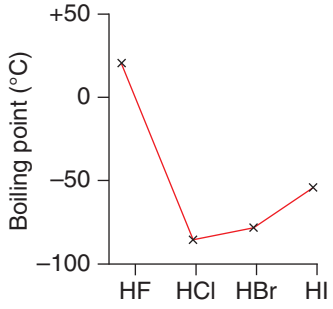


Fig. 24.17 The boiling points of the hydrogen halides

If hydrogen fluoride only had van der Waals' forces between its molecules, we would expect its boiling point to be about –88 °C. However, the boiling point of hydrogen fluoride is much higher. There must be some additional intermolecular forces of attraction which requires significantly more heat to break. These relatively powerful intermolecular forces are described as **hydrogen bonds**. Hydrogen bonds form when hydrogen is covalently bonded to the very electronegative elements fluorine, oxygen or nitrogen.

When a hydrogen atom is covalently bonded to a very electronegative fluorine atom, a very polar H–F covalent bond is formed. The hydrogen atom has a strong partial positive charge.

Fig. 24.18a shows the hydrogen bonding in liquid hydrogen fluoride. In a hydrogen fluoride molecule, the fluorine atom strongly attracts the shared pair of electrons in the covalent bond towards itself. This causes the hydrogen atom's nucleus of a single proton to be exposed. The proton is attracted strongly to a lone pair of electrons on the fluorine atom of another hydrogen fluorine molecule. The resulting force of attraction is a hydrogen bond.

DSE 2012 Paper 1B Q4

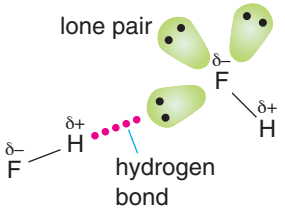


Fig. 24.18a The hydrogen bond between the partially positive hydrogen atom and the lone pair on the fluorine atom of another molecule

hydrogen bond 氫鍵

✓ A hydrogen bond is the particularly strong attractive force between the hydrogen atom attached to an electronegative atom and the lone pair on another electronegative atom. Usually the electronegative atom is fluorine, oxygen or nitrogen.

Hydrogen bonds are very strong compared with other intermolecular forces. This means that for hydrogen fluoride to boil, these strong hydrogen bonds have to be broken before the molecules can change from liquid state into gaseous state. This results in the unusually high boiling point for hydrogen fluoride.

Notice that in liquid hydrogen fluoride, each molecule can be hydrogen-bonded to two other molecules (Fig. 24.18b).

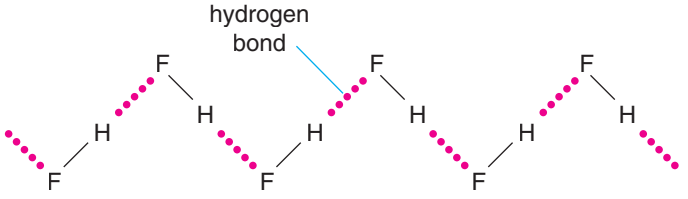


Fig. 24.18b In liquid hydrogen fluoride, each molecule can be hydrogen-bonded to two other molecules

Reference Website
Refer to the following websites for animations showing the hydrogen bonds between water molecules:
- http://www.northland.cc.mn.us/biology/Biology1111/animations/hydrogenbonds.html (accessed July 2014)
- http://dwb4.unl.edu/ChemAnime/HYBOND/HYBONDD.html (accessed July 2014)

The hydrogen bonding in ammonia and water is shown in Fig. 24.19.

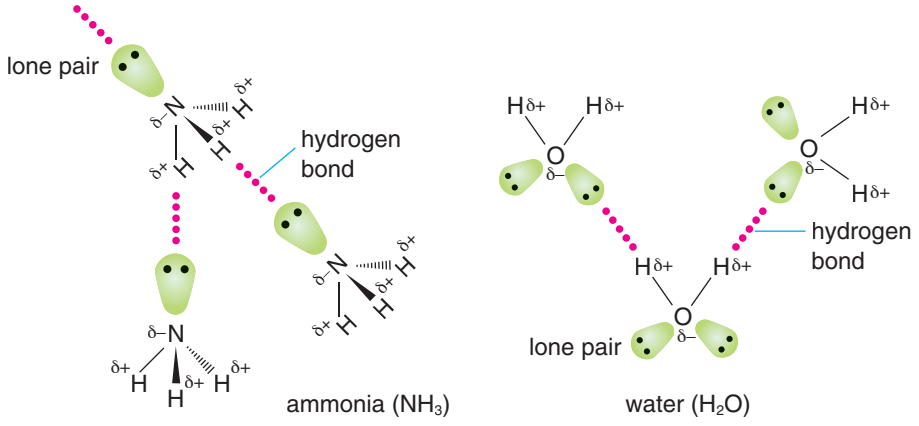


Fig. 24.19 Hydrogen bonding in ammonia and water

Notice that the angle between the covalent bond linking the hydrogen atom and the hydrogen bond is usually 180°.

Other electronegative elements such as chlorine and sulphur do not form hydrogen bonds like fluorine, oxygen or nitrogen does. Although H–Cl and H–S bonds are polar, the attractions between the hydrogen atoms and the lone pairs on other chlorine and suphur atoms are not as strong.

Relative strengths of intermolecular forces

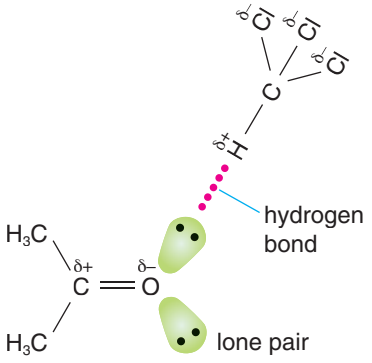
Covalent bond strengths are typically within 200–500 kJ mol^{-1}, although they can be outside this range. Hydrogen bonds are weak in comparison — about 5–40 kJ mol^{-1}. Van der Waals' forces are weaker still.

Example 24.2

Q Draw a diagram to show the hydrogen bonding between molecules of propanone (CH_3COCH_3) and trichloroethane ($CHCl_3$). Explain why hydrogen bonding forms between the molecules.

A The trichloromethane molecule is very polar. Its hydrogen atom has a partial positive charge. Hydrogen bonding forms between this hydrogen atom and the lone pair on the highly electronegative oxygen atom of a propanone molecule.

We have discussed the polar trichloromethane molecule earlier in this unit.



Example 24.3

Q Account for the following:

$NH_3(g)$ is very soluble in water, but $PH_3(g)$ is almost insoluble.

ASL 2010 Paper 2 Q5(b)
AL 2010 Paper 2 Q3(c)(ii)

A NH_3 molecules can form hydrogen bonds with water molecules extensively.

The electronegativity of P is very close to that of H. PH_3 molecule has a very small dipole moment. Its interaction with water molecules is much weaker than the hydrogen bonds between water molecules. Thus, PH_3 cannot mix with water.

Practice 24.4

1 Predict the type(s) of intermolecular forces between molecules of:

a) chloromethane (CH_3Cl);
Van der Waals' forces

b) methanol (CH_3OH).
Hydrogen bonds and van der Waals' forces

(Remark: More than one type of intermolecular forces may operate for each liquid.)

2 b) Consider an NH_3 molecule. When a hydrogen atom is covalently bonded to a very electronegative nitrogen atom, a very polar H–N covalent bond is formed. The hydrogen atom has a strong partial positive charge.

The hydrogen atom is attracted strongly to a lone pair of electrons on the nitrogen atom of another NH_3 molecule. The resulting force of attraction is a hydrogen bond.

The boiling point of a compound depends on the strength of its intermolecular attractions.

Hydrogen bonds exist in NH_3 while only van der Waals' forces exist in PH_3.

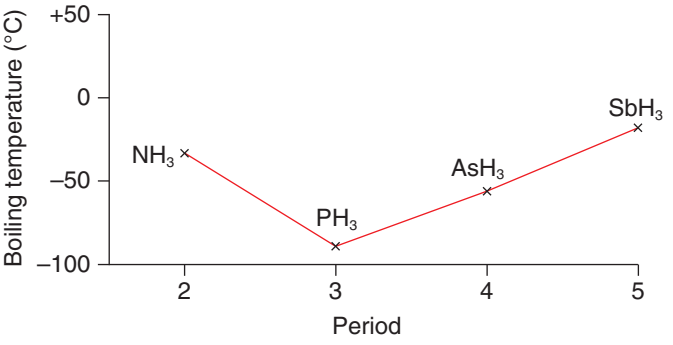
Hydrogen bonds are stronger than van der Waals' forces. More heat is needed to separate the NH_3 molecules during boiling.

Thus, the boiling point of NH_3 is higher than that of PH_3.

The intermolecular attractions in PH_3, AsH_3 and SbH_3 are van der Waals' forces.

The number of electrons in one molecule increases from PH_3 to SbH_3. Hence the strength of van der Waals' forces also increases from PH_3 to SbH_3. This suggests that the boiling points should increase from PH_3 to SbH_3, in accordance with the data.

2 The graph below shows the boiling points of hydrides of Group V elements.



a) Describe the trend in boiling points from NH_3 to SbH_3. The boiling point of NH_3 is higher than that of PH_3. The boiling point increases from PH_3 to SbH_3.

b) Explain the trend in terms of intermolecular forces and electronegativities.

24.10 The structure and density of ice

If we take a tube of water at room temperature and cool it, the water contracts and its density increases until the temperature reaches 4 °C (Fig. 24.20). Water reaches its maximum density at 4 °C. As it cools further and freezes into ice, it actually becomes less dense.

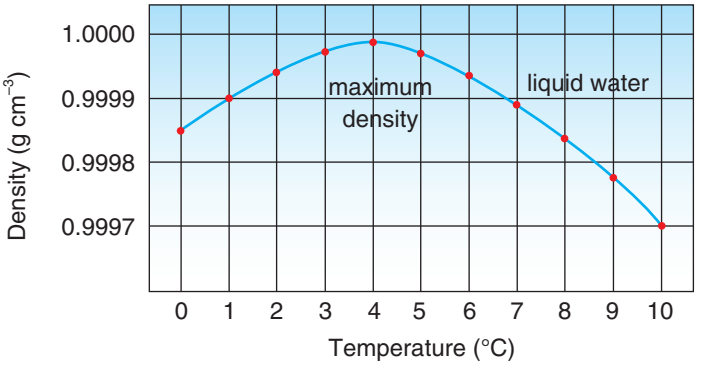


Fig. 24.20 A plot of density against temperature for liquid water

DSE 2013 Paper 1B Q1(c)
AL 2006 Paper 2 Q2(b)(i)

Compare the arrangement of water molecules in water and ice (Fig. 24.21). In ice, each molecule forms four hydrogen bonds tetrahedrally. The highly ordered structure of ice prevents molecules from getting too close to one another and leads to a very 'open' structure with large spaces in it. In water, the molecules have relative motion and are more closely packed. Hence the water molecules in ice take up more volume than those in water; ice is less dense than water. N6 This explains why ice floats on the surface of water (Fig. 24.22).

N6 Examination questions often ask students to explain why ice is less dense than water.

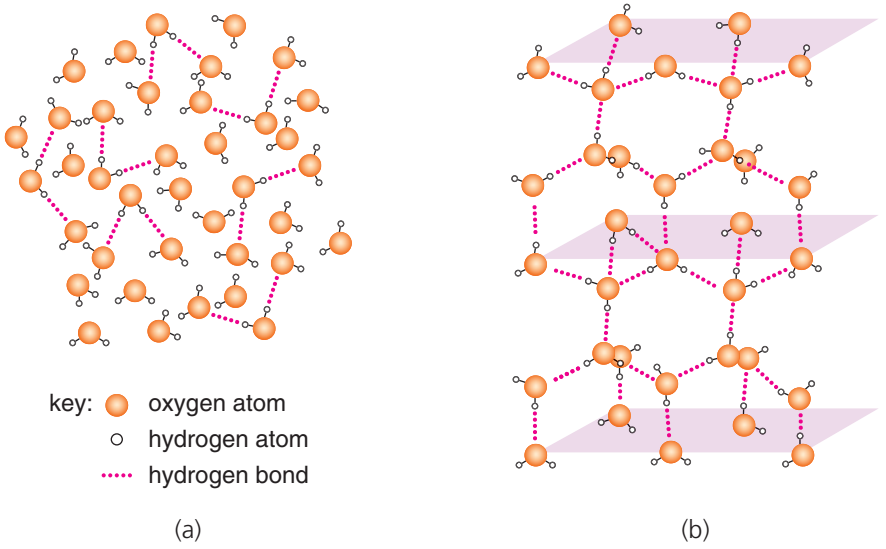


Fig. 24.21 The arrangement of water molecules in (a) water and (b) ice



Fig. 24.22 A polar bear is jumping between ice pieces which float on the surface of water

When water is cooled to 4 °C, its molecules begin to adopt the open structure of ice. This explains why the fall in density begins as water is cooled to 4 °C.

Video
Investigating the surface tension and viscosity of water

24.2
Investigating the surface tension and viscosity of water.

24.11 Surface tension and viscosity of liquids

Intermolecular forces give rise to a number of structural features and physical properties of liquids. In this section, we will look at two such properties: **surface tension** and **viscosity**.

Surface tension

Video
Surface tension of water and using soap or detergent as a wetting agent

Molecules within a liquid are pulled in all directions by intermolecular forces. However, molecules at the liquid surface are subject to attractions only from the side and from below (Fig. 24.23). The effect of this uneven pull on the surface molecules draws them into the body of the liquid, causing the surface to tighten like an elastic membrane. Thus, the surface is under tension.

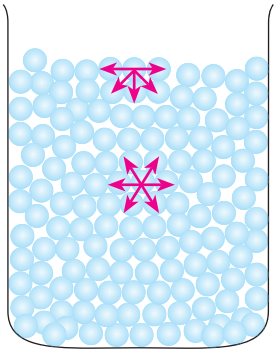


Fig. 24.23 Molecules at the liquid surface are subject to attractions only from the side and from below



Fig. 24.24 A pond skater is walking across the surface of a pond

Surface tension could be defined as the property of the surface of a liquid that allows it to resist an external force, due to the intermolecular forces between the liquid molecules. Liquids that have strong intermolecular forces tend to have high surface tension.

Because of surface tension, a small insect like a pond skater can walk across the surface of a pond (Fig. 24.24), and an object of greater density than water can be made to float (Fig. 24.25).



Fig. 24.25 A coin of greater density than water can float on the water surface

surface tension 表面張力　　viscosity 黏度



Fig. 24.26 Less viscous liquids like water flow freely when poured (left), whilst more viscous liquids like motor oil flow slowly (right)

Viscosity

Reference Website
Refer to the following website for an animation comparing the flow rate of two liquids with different viscosity:
http://plc.cwru.edu/tutorial/enhanced/lab/visco/intro/intro.htm
(accessed July 2014)

The viscosity of a liquid is a measure of a liquid's resistance to flow. The greater is the viscosity, the more slowly the liquid flows (Fig. 24.26).

The viscosity of a liquid depends on:

- the strength of attractive forces between molecules; and
- the tendency of molecules to become *entangled* with each other.

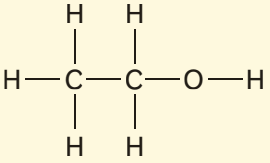
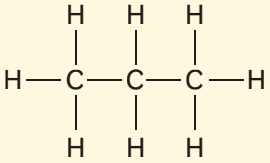
A liquid with strong intermolecular forces has a higher viscosity than one with weak intermolecular forces. If the molecules of a liquid are in the form of long chains, they can become entangled and cannot slide past one another easily. Thus, the viscosity of the liquid is high.

24.12 Intermolecular attractions and properties of alcohols

Table 24.2 lists the types of intermolecular attractions existing in **ethanol** and **propane**, as well as their boiling points and solubilities in water.

Table 24.2

Types of intermolecular attractions existing in ethanol and propane

Substance	Relative molecular mass	Structure	Type(s) of intermolecular attractions present		Boiling point (°C)	Solubility in water
			van der Waals' forces	Hydrogen bonds		
Ethanol	46.0		✔	✔	78	soluble in all proportions
Propane	44.0		✔		–42	insoluble

entangled 使纏繞　　ethanol 乙醇　　propane 丙烷

Boiling points of ethanol and propane

24.3

Determining the strength of the hydrogen bond between ethanol molecules.

The boiling point of ethanol is much higher than that of propane. This is due to the presence of hydrogen bonds between ethanol molecules (Fig. 24.27) in addition to van der Waals' forces. More heat is needed to separate the ethanol molecules during boiling.

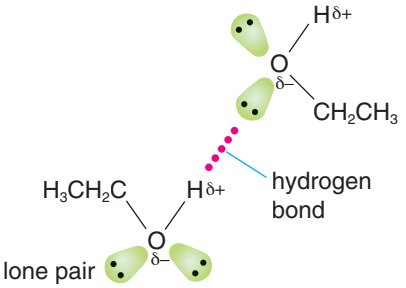


Fig. 24.27 Hydrogen bonding in ethanol

Example 24.4

Q An experiment is carried out to determine the strength of the hydrogen bond formed between molecules of trichloromethane ($CHCl_3$) and ethyl ethanoate ($CH_3COOC_2H_5$).

14.9 g of trichloromethane were mixed with 18.0 g of ethyl ethanoate. A maximum temperature rise of 6.00 °C was recorded.

ASL 2011 Paper 1 Q3(a)
AL 2011 Paper 1 Q3(a)(i)

You are provided with the following data:

Compound	Trichloromethane	Ethyl ethanoate
Molar mass (g mol^{-1})	119.5	88.0
Specific heat capacity (J g^{-1} K^{-1})	0.970	1.92

a) Draw a diagram to show the hydrogen bond formed between molecules of trichloromethane and ethyl ethanoate.

b) Deduce which compound is in excess. Why should an excess of this compound be used?

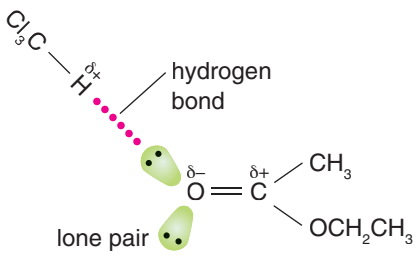
c) i) Calculate the amount of heat required to raise the temperature of trichloromethane in the experiment.

ii) Calculate the amount of heat required to raise the temperature of ethyl ethanoate in the experiment.

iii) Using the values from (i) and (ii), calculate the strength of hydrogen bond (in kJ mol^{-1}) between molecules of trichloromethane and ethyl ethanoate.

Continued on next page

A1 a)



b) Number of moles of $CHCl_3$

$= \dfrac{14.9 \text{ g}}{119.5 \text{ g mol}^{-1}}$

$= 0.125 \text{ mol}$

Number of moles of $CH_3COOC_2H_5$

$= \dfrac{18.0 \text{ g}}{88.0 \text{ g mol}^{-1}}$

$= 0.205 \text{ mol}$

$\therefore$ $CH_3COOC_2H_5$ is in excess.

As $CH_3COOC_2H_5$ is in excess, we can assume that each $CHCl_3$ molecule forms one hydrogen bond with one $CH_3COOC_2H_5$ molecule.

c) i) Amount of heat required to raise the temperature of trichloromethane

= mass x specific heat capacity x rise in temperature

$= 14.9 \text{ g} \times 0.970 \text{ J g}^{-1} \text{ K}^{-1} \times 6.00 \text{ K}$

$= 86.7 \text{ J}$

We will further discuss this formula in Topic 9 Chemical Reactions and Energy.

ii) Amount of heat required to raise the temperature of ethyl ethanoate

= mass x specific heat capacity x rise in temperature

$= 18.0 \text{ g} \times 1.92 \text{ J g}^{-1} \text{ K}^{-1} \times 6.00 \text{ K}$

$= 207 \text{ J}$

A 1 °C interval on the Celsius scale is equivalent to an interval of 1 kelvin (symbol: K) on the absolute temperature scale.

iii) Amount of heat released due to the formation of hydrogen bond

$= (86.7 + 207) \text{ J}$

$= 294 \text{ J}$

Strength of hydrogen bond between molecules of trichloromethane and ethyl ethanoate

$= \dfrac{294 \text{ J}}{0.125 \text{ mol}}$

$= 2\,350 \text{ J mol}^{-1}$

$= 2.35 \text{ kJ mol}^{-1}$

$\therefore$ the strength of hydrogen bond between molecules of trichloromethane and ethyl ethanoate is 2.35 kJ mol^{-1}.

DSE 2014 Paper 1B Q2
ASL 2011 Paper 1 Q4(a)
AL 2011 Paper 1 Q3(b)(i)

Solubility of ethanol and propane in water

N7 Ethane-1,2-diol ($HOCH_2CH_2OH$) is soluble in water because
- it has a small molecular size / it has a short carbon chain;
- the hydroxyl groups in it can form hydrogen bonds with water molecules.

Ethanol is soluble in water because hydrogen bonds can form between ethanol molecules and water molecules (Fig. 24.28). This helps the dissolving process. N7

Propane is non-polar and not miscible with water. The strength of intermolecular attractions between water molecules is greater than that between water and propane molecules. Thus, propane is insoluble in water.

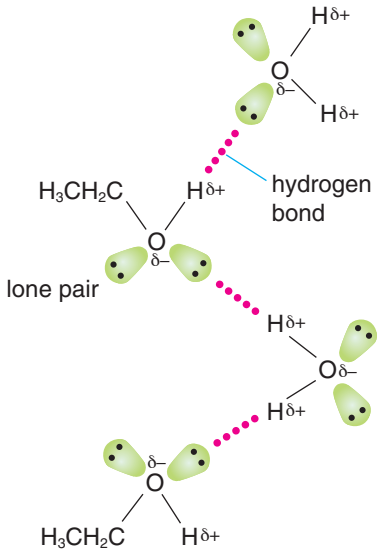
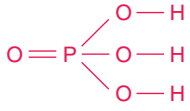


Fig. 24.28 The hydrogen bonding between ethanol molecules and water molecules

Viscosity of alcohols

N8 Concentrated H_3PO_4 has a high viscosity due to the strong hydrogen bonding between its molecules.

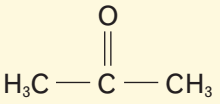
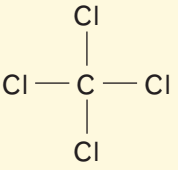
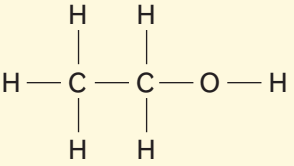
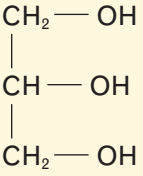


The viscosity of ethanol is much higher than those of many other liquids because of its ability to form hydrogen bonds. How does the extent of hydrogen bonding in alcohols with different numbers of hydroxyl groups (–OH) affect the viscosity of the alcohols? N8

Table 24.3 lists the relative viscosity of some common liquids at 25 °C.

24.4
Comparing the viscosity of alcohols with different numbers of hydroxyl groups.

Table 24.3
The relative viscosity of some common liquids at 25 °C

Liquid	Structure	Relative viscosity at 25 °C
Propanone		*
Tetrachloromethane		*
Ethanol		**
Glycerol		*****

(***** = highest, * = lowest)

The viscosity of glycerol is much higher than those of all the other liquids listed in Table 24.3. Each glycerol molecule has three –OH groups that can take part in hydrogen bonding with other glycerol molecules (Fig. 24.29). Furthermore, because of their shape, the molecules tend to become entangled rather than to slide past one another. These factors contribute to the high viscosity of glycerol.

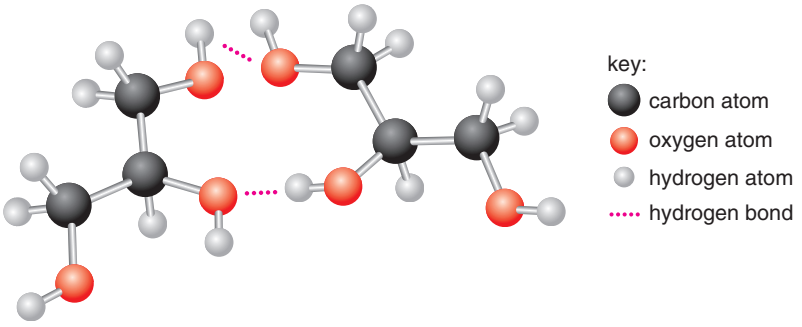


Fig. 24.29 Glycerol has a high viscosity because of its ability to form multiple hydrogen bonds

glycerol 甘油

Example 24.5

Q The table below lists the boiling points and densities of three alcohols.

Alcohol	Structural formula	Boiling point (°C)	Density at 20 °C (g cm^{-3})
A	CH_3CH_2OH	78	0.789
B	$CH_3CH_2CH_2OH$	97	0.804
C	$CH_3CH(OH)CH_3$	82	0.786

a) Explain why the boiling point of alcohol B is higher than that of alcohol A.

b) The density of alcohol B is higher than that of alcohol C. Explain why.

A a) The boiling point of a compound depends on the strength of its intermolecular attractions.

Both alcohols have a −OH group. Their boiling point difference is related to the difference in the strength of van der Waals' forces between molecules. The strength of van der Waals' forces in the alcohols increases with the number of electrons in a molecule.

Compared with alcohol A, alcohol B has a greater number of electrons in a molecule. The van der Waals' forces in B are stronger than those in A.

Thus, the boiling point of alcohol B is higher than that of alcohol A.

b) For alcohols with the same molecular formula, the strength of intermolecular forces is affected by the shape of molecules.

Molecule of alcohol B has a linear shape while that of alcohol C has a more spherical and compact shape. The molecular shape of alcohol B allows greater surface contact between molecules.

The van der Waals' forces in alcohol B are stronger, pulling the molecules closer together.

So, the density of alcohol B is higher.

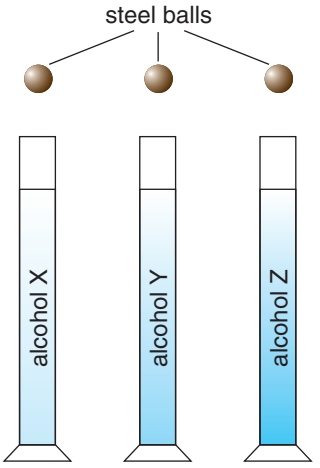
Discussion

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

The structural formula of three alcohols, X, Y and Z, are shown below.

Alcohol	Structural formula
X	$CH_3CH_2CH_2OH$
Y	$HOCH_2CH_2CH_2OH$
Z	$HOCH_2CH(OH)CH_2OH$

In an experiment, three identical steel balls are dropped through alcohols X, Y and Z.



Which ball would take the longest time to reach the bottom? Discuss.

The rate at which each steel ball drops depends on the viscosity of the alcohol. This is related to the intermolecular attractions existing in the liquid.

Alcohol X has one, alcohol Y has two and alcohol Z has three –OH group(s) per molecule that can take part in hydrogen bonding. Thus, the intermolecular attractions in alcohol Z is the strongest.

Among the three alcohols, the viscosity of alcohol Z is the highest because it has the strongest intermolecular attractions and its molecules tend to become entangled due to the shape.

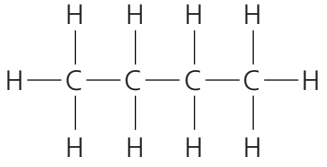
Thus, the ball drops most slowly in alcohol Z.

Practice 24.5

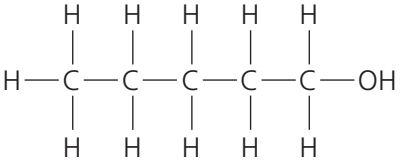
Refer to page T18.

Consider the following four compounds, A, B, C and D:

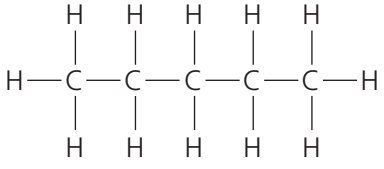
A



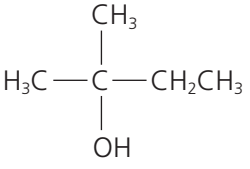
B



C



D



Arrange the four compounds in order of increasing boiling point. Explain your answer.

Have you mastered?

Key terms

non-polar 非極性 25 polar 極性 25 electronegativity 電負性 26 dipole 偶極 28
dipole moment 偶極矩 28 intermolecular force 分子間引力 34 instantaneous dipole 瞬間偶極 35
induced dipole 誘發偶極 35 permanent dipole 永久偶極 36
van der Waals' forces 范德華力 36 hydrogen bond 氫鍵 40 surface tension 表面張力 45
viscosity 黏度 45 ethanol 乙醇 46 propane 丙烷 46

Checklist

After studying this unit, you should be able to

- ☐ define the electronegativity of an element;
- ☐ identify the partial charges of polar molecules;
- ☐ explain the non-polar nature of CO_2, BF_3 and CCl_4;
- ☐ explain the existence of van der Waals' forces in non-polar and polar covalent substances;
- ☐ describe factors that affect the strength of van der Waals' forces;
- ☐ describe the formation of hydrogen bonding;
- ☐ compare the strength of van der Waals' forces with that of hydrogen bonding;
- ☐ understand the effect of hydrogen bonding on properties of substances such as water and ethanol;
- ☐ describe the structure of ice;
- ☐ state and explain the density of ice in terms of its structure and bonding.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

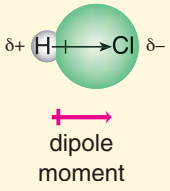
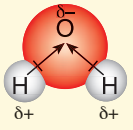
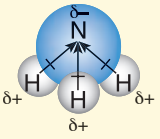
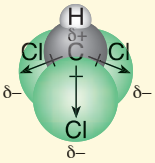
1 Polar covalent bonding occurs because one bonded atom has a stronger attraction for electrons than the other bonded atom does.

2 The electronegativity of an element represents the power of an atom of that element to attract a bonding pair of electrons towards itself in a molecule.

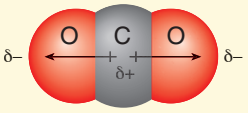
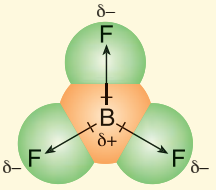
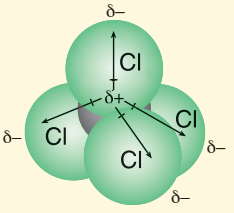
3 For a molecule to be polar, the following conditions must apply:

- the molecule must contain one or more polar bonds;
- the shape of the molecule must not be completely symmetrical.

4 The following table shows some polar molecules.

Molecule	HCl	H_2O	NH_3	$CHCl_3$
Shape				

5 The following table shows some non-polar molecules.

Molecule	CO_2	BF_3	CCl_4
Shape			

The molecules are non-polar because the identical bond dipole moments cancel one another out exactly. The molecules have no net dipole moment.

6 a) Van der Waals' forces are stronger between molecules with

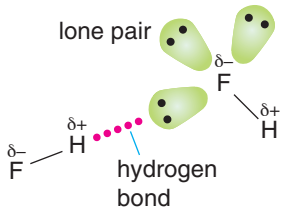
- a greater number of electrons;
- a greater relative molecular mass;
- a greater molecular size.

b) Van der Waals' forces are stronger between molecules with a more spread-out shape (i.e. a greater molecular surface area).

Have you mastered?

7 A hydrogen bond is the particularly strong attractive force between the hydrogen atom attached to an electronegative atom and the lone pair on another electronegative atom. Usually the electronegative atom is fluorine, oxygen or nitrogen.

The following diagram shows the hydrogen bonding between hydrogen fluoride molecules.



Fluorine is a very electronegative element.

The H–F covalent bond is very polar. The hydrogen atom has a strong partial positive charge.

Hydrogen bonding exists between the hydrogen atom of one molecule and a lone pair on the fluorine atom of another molecule.

8 In the structure of ice, each water molecule forms four hydrogen bonds tetrahedrally. This leads to a very 'open' structure with large spaces in it. Thus, the ice is less dense than liquid water.

9 Intermolecular forces give rise to a number of structural features and physical properties of liquids.

a) A liquid with strong intermolecular forces tends to have a high surface tension.

b) The viscosity of a liquid is a measure of a liquid's resistance to flow.

The viscosity of a liquid depends on:

- the strength of attractive forces between molecules; and
- the tendency of molecules to become entangled with each other.

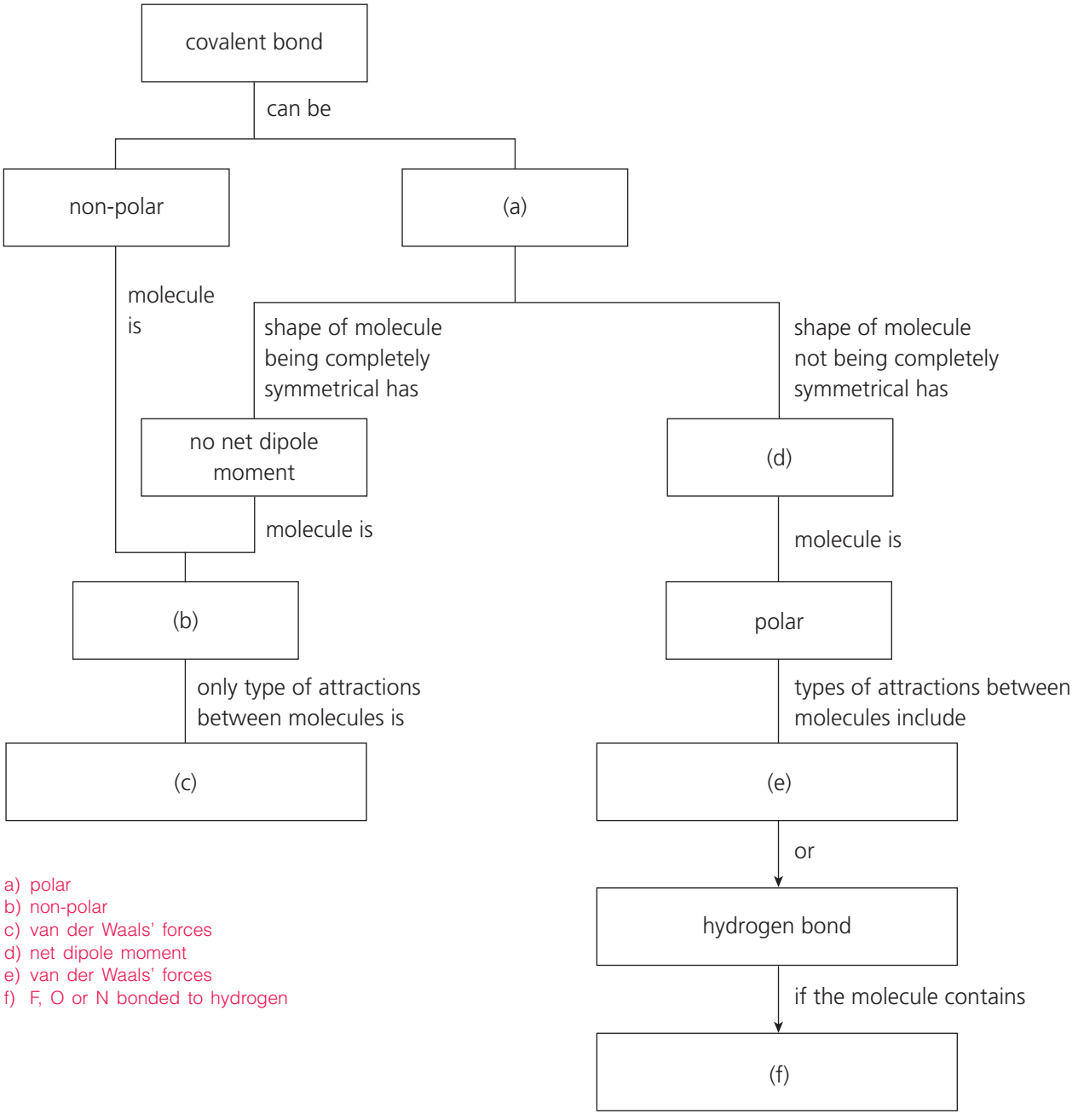
10 Ethanol is soluble in water because hydrogen bonds can form between ethanol molecules and water molecules.

Unit Exercise

Note: The symbol 🌶 indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the following concept map.



a) polar
b) non-polar
c) van der Waals' forces
d) net dipole moment
e) van der Waals' forces
f) F, O or N bonded to hydrogen

2 a) The electronegativity of an element represents the power of an atom of that element to attract a bonding pair of electrons towards itself in a molecule.

2 a) Explain the meaning of the term 'electronegativity'.

b) The electronegativity values of some elements are given below.

Element	H	C	N	O
Electronegativity	2.1	2.5	3.0	3.5

Mark, using the symbol δ+ and δ–, the partial charges which exist in the covalent bonds below.

$\overset{\delta-}{C}–\overset{\delta+}{H}$ $\overset{\delta+}{C}=\overset{\delta-}{O}$ $\overset{\delta-}{O}–\overset{\delta+}{H}$ $\overset{\delta-}{N}–\overset{\delta+}{H}$

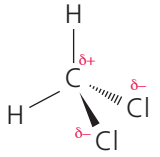
c) From the following compounds:

CH_4 CO_2 H_2O NH_3

choose the compound which contains the least polar bond.

3 The molecule CH_2Cl_2 has a tetrahedral shape. The C–Cl bonds are polar.

a) Mark partial charges (δ+ and δ–) on the C and Cl atoms in the structure below.



b) Explain what determines where the partial charges are placed on this molecule.

c) Does the whole molecule have a net dipole moment? Explain your answer.

b) The difference in electronegativity values of carbon and chlorine determine where the partial charges are placed on the molecule.

c) Yes.
Each C–Cl bond is polar.
Because of its tetrahedral shape, the individual C–Cl bond dipole moments reinforce each other.
Hence the whole molecule has a net dipole moment.

Part II Multiple choice questions

4 Which of the following elements is the most electronegative?

A Argon
B Carbon
C Fluorine
D Nitrogen

5 Tetrachloromethane, CCl_4, is a
Refer to page T19 for explanation.
A polar molecule with polar bonds.
B polar molecule with non-polar bonds.
C non-polar molecule with polar bonds.
D non-polar molecule with non-polar bonds.

(Edexcel Advanced Subsidiary GCE, Unit 2, Jan. 2012, 2)

6 Which one of the following molecules has a zero dipole moment?

A BF_3
B PH_3
C SO_2
D HCl

(HKASLE, Paper 1, 2008, 5(b))

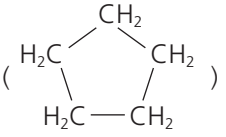
7 Which of these four molecules, PCl_3, CO, CO_2 and CCl_4, are polar?

A All four
B PCl_3 and CO
C CO and CCl_4
D PCl_3 and CO_2

CO_2 and CCl_4 molecules contain polar bonds. However, they have symmetrical shapes. The bond dipole moments cancel one another out exactly. Thus, these molecules are non-polar.

(Edexcel Advanced Subsidiary GCE, Unit 2, Jan. 2009, 7)

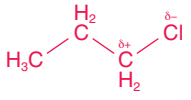
8 An electric field can affect the direction of a stream of some liquids. Which of these liquids would be affected by an electric field?

A 1-chloropropane ($CH_3CH_2CH_2Cl$)
B Pentane ($CH_3CH_2CH_2CH_2CH_3$)
C Tetrachloromethane (CCl_4)
D Cyclopentane (

)

(Edexcel Advanced Subsidiary GCE, Unit 2, Jun. 2009, 16)

1-chloropropane molecules are polar.



Thus, a stream of 1-chloropropane would be affected by an electric field.

9 Hydrogen chloride has a low boiling point because

A weak covalent bonds exist between hydrogen chloride molecules.
B weak covalent bonds exist between hydrogen atoms and chlorine atoms.
(C) weak van der Waals' forces exist between hydrogen chloride molecules.
D weak van der Waals' forces exist between hydrogen ions and chloride ions.

(HKCEE, Paper 2, 2011, 5)

10 Which of the following compounds shows hydrogen bonding in the liquid state?

A Hydrogen bromide, HBr
B Hydrogen sulphide, H_2S
C Silane, SiH_4
(D) Ammonia, NH_3

(Edexcel Advanced Subsidiary GCE, Unit 2, Jun. 2010, 7)

11 The following liquids all have the same number of electrons in each molecule. Which one is likely to have the lowest boiling point?

A $CH_3CH_2CH_2CH_2OH$
B $CH_3CH_2CH_2CH_2CH_3$
(C) $CH_3C(CH_3)_2CH_3$
D $CH_3CH(CH_3)CH_2CH_3$

(Edexcel Advanced Subsidiary GCE, Unit 2, Jan. 2009, 9)

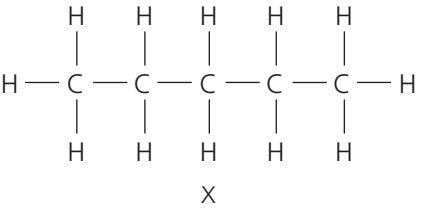
12 Ammonia is very soluble in water. Which of the following statements best accounts for this phenomenon?

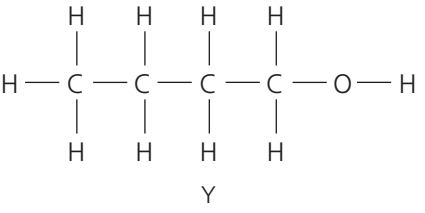
A Both ammonia molecule and water molecule are polar.
B Ammonia molecule and water molecule are of comparable sizes.
C Ammonia undergoes ionization in water.
(D) Ammonia forms hydrogen bond with water.

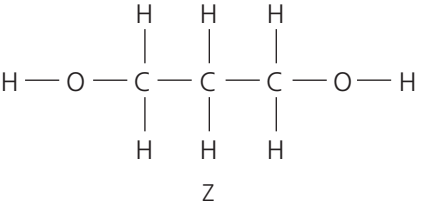
(HKDSE, Practice paper 1A, 2012, 17)

11 Option A — Hydrogen bonds exist in $CH_3CH_2CH_2CH_2OH$ while only van der Waals' forces exist in the other three compounds. Thus, the boiling point of $CH_3CH_2CH_2CH_2OH$ is the highest.
Option C — The $CH_3C(CH_3)_2CH_3$ molecule is spherical and compact. The molecular surface area available for coming into contact with neighbouring molecules is the smallest. Thus, the van der Waals' forces between $CH_3C(CH_3)_2CH_3$ molecules are weakest.

13 Consider the following three compounds:
Refer to page T19 for explanation.


X


Y


Z

Which of the following sequences correctly lists the compounds in order of increasing water solubility?

A Z < Y < X
B Y < Z < X
C X < Z < Y
(D) X < Y < Z

14 Which of the following statements about electronegativity are correct?

(1) Electronegativity increases from left to right across the second period of the periodic table.
(2) The electronegativity of Group I elements increases with increasing atomic number.
(3) The electronegativity of Group VII elements decreases with increasing atomic number.

A (1) and (2) only
(B) (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(2) The electronegativity of Group I elements decreases with increasing atomic number.

Unit Exercise

15 Compounds X and Y have the same molecular formula. Their structures are shown below.

Compound X $CH_3CH_2CH_2OH$

Compound Y $CH_3CH(OH)CH_3$

Which of the following statements about compounds X and Y are correct?

(1) Each can form hydrogen bond with another of its own molecule.
(2) The boiling point of compound X is higher than that of compound Y.
(3) The density of compound Y is higher than that of compound X.

(A) (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

15 (2) Both alcohols have a –OH group. Their boiling point difference is due to the difference in the strength of van der Waals' forces between molecules.

Molecule of X has a linear shape while that of Y has a more spherical and compact shape. The molecular shape of X allows greater surface contact between molecules. The van der Waals' forces in X are stronger. Thus, the boiling point of X is higher than that of Y.

(3) The van der Waals' forces in X are stronger, pulling the molecules close together. Thus, the density of X is higher than that of Y.

Part III Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

16 Linus Pauling was a Nobel prize winning chemist who devised a scale of electronegativity.

Some Pauling electronegativity values are shown in the table.

Element	Electronegativity
B	2.0
Br	2.8
N	3.0
F	4.0

a) Show, using δ+ and δ– symbols, the permanent dipoles on each of the following bonds.

$\overset{\delta+}{N}–\overset{\delta-}{F}$ $\overset{\delta-}{N}–\overset{\delta+}{Br}$

b) Boron trifluoride (BF_3), ammonia (NH_3) and sulphur hexafluoride (SF_6) are all covalent compounds. The shapes of their molecules are different.

i) State the shape of a molecule of SF_6.
Octahedral

ii) Using *outermost electron shells* only, draw electron diagrams for molecules of BF_3 and NH_3.

16 b) ii) Molecule of BF_3 Molecule of NH_3



iii) BF_3 has a symmetrical shape.
The three identical B–F bond dipole moments cancel one another out exactly.

iii) Molecules of BF_3 contain polar bonds, but the molecules are non-polar.

Suggest an explanation for this difference.

(OCR Advanced Subsidiary GCE, Chem. A, F321, Jan. 2011, 3(b)–(c))

17 Snottites are slimy colonies of bacteria that live in extreme conditions in caves. They take in hydrogen sulphide gas, H_2S, and oxygen gas and form sulphuric acid, H_2SO_4, in their slime.
Refer to page T19.

a) Hydrogen sulphide is produced by sulphur springs and stagnant water.

i) Draw an electron diagram to represent the bonding in a molecule of hydrogen sulphide.

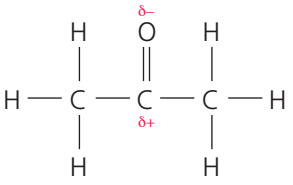
Show *outermost electron shells* only.

ii) Suggest and explain the shape of the hydrogen sulphide molecule.

b) Use your answer to part (a)(ii) to help explain whether or not the molecule of hydrogen sulphide is polar.

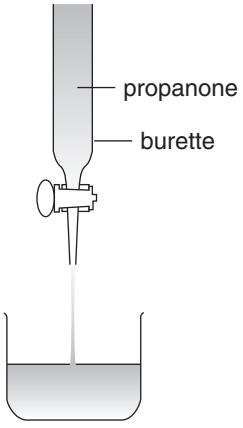
(OCR Advanced Subsidiary GCE, Chem. B (Salters), F332, Jan. 2012, 1(a)(ii), (b))

18 The structure of propanone is shown below:

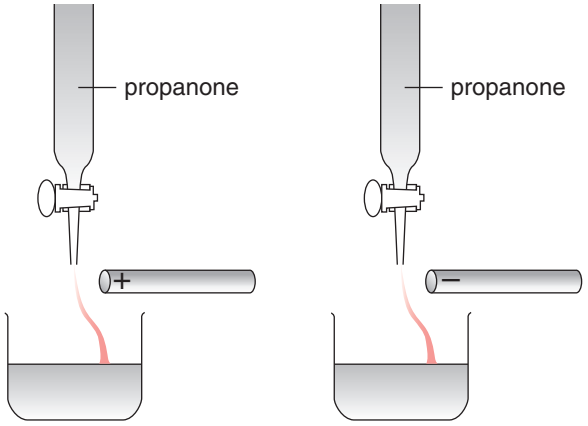


a) The C=O bond in propanone is polar. Mark the partial charges (δ+ and δ–) on the atoms of the bond.

b) A jet of propanone is run from a burette as shown below:



A positively charged rod and a negatively charged rod are placed near the jet of propanone as shown in diagrams (i) and (ii) below:



(i) (ii)

Show and explain your expected observation in each case.

18 b) i) The jet of propanone is deflected by the positively charged rod.
The negative ends of the molecules are attracted towards the rod.

ii) The jet of propanone is deflected by the negatively charged rod.
The positive ends of the molecules are attracted towards the rod.

19 The table shows the boiling points of four halogens placed in order of increasing atomic number.

Refer to page T19.

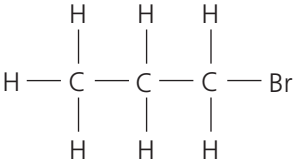
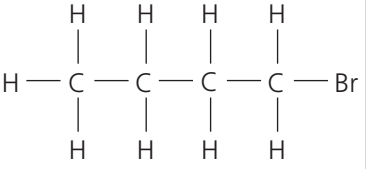
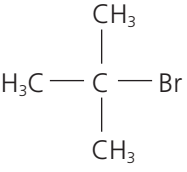
Halogen	Boiling point (°C)
Fluorine, F_2	–188
Chlorine, Cl_2	–34
Bromine, Br_2	59
Iodine, I_2	184

Explain, in terms of the variation in intermolecular forces, why the boiling points increase from fluorine to iodine.

(Edexcel GCSE (Higher Tier), Chemistry, Unit C2, Mar. 2012, 4(d))

20 The table below lists the boiling points of three compounds, X, Y and Z.

Refer to page T19.

Compound	Structure	Boiling point (°C)
X		71.1
Y		101.7
Z		73.4

Explain why

a) compound Y has a higher boiling point than compound X;

b) compound Y has a higher boiling point than compound Z.

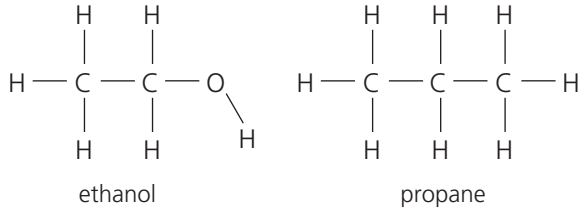
22 a) i) Type of bond: Covalent bond
Explanation: One pair of outermost shell electrons are shared between the C and H atoms.

ii) Liquid ethanol: Hydrogen bonds
Liquid propane: Van der Waals' forces

21 With the aid of a diagram, explain the formation of hydrogen bonding in hydrogen fluoride.

(HKDSE, Paper 1B, 2012, 4)

22 a) Two carbon compounds with similar relative molecular masses are shown below.

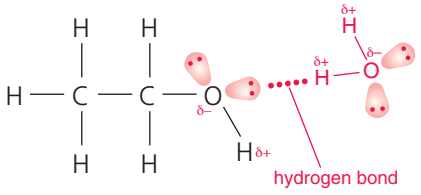


ethanol propane

i) State the type of bond present between the C and H atoms in both of these molecules. Explain how this type of bond is formed.

ii) State the strongest type of intermolecular force present in each compound.

b) Ethanol dissolves in water. Draw a diagram to show how one molecule of ethanol interacts with one molecule of water in the solution. Include partial charges and all lone pairs. The ethanol molecule has been drawn for you.



(AQA Advanced Subsidiary GCE, Unit 1, Jun. 2009, 3(a)–(b))

23 A group of students conducted an experiment to investigate the strength of hydrogen bonds between ethanol molecules.

They mixed 8.0 g of ethanol with 16.0 g of cyclohexane (excess). A maximum temperature fall of 3.8 K was recorded.

The following information was provided:

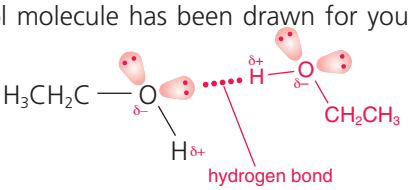
Compound	Ethanol	Cyclohexane
Specific heat capacity (J g^{-1} K^{-1})	2.41	1.83

Amount of heat taken in by a substance
= mass x specific heat capacity x fall in temperature

Molar mass of ethanol = 46.0 g mol^{-1}

a) Draw a diagram to show the hydrogen bond formed between two ethanol molecules. Include partial charges and all lone pairs.

One ethanol molecule has been drawn for you.



b) What happened to the hydrogen bonds between ethanol molecules on mixing ethanol with cyclohexane?
Hydrogen bonds between ethanol molecules were broken.

c) i) Calculate the amount of heat taken in by ethanol in the experiment.
73.3 J; refer to page T20.

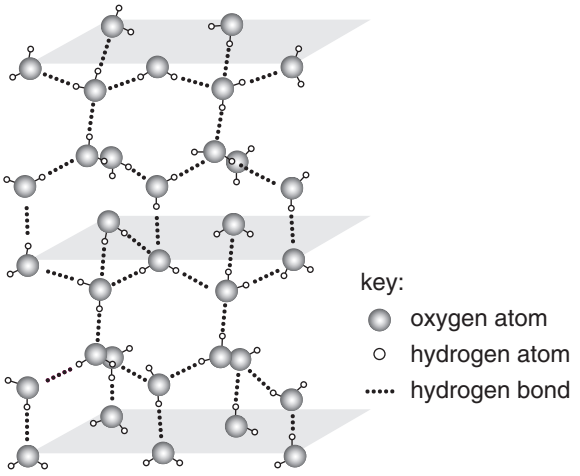
ii) Calculate the amount of heat taken in by cyclohexane in the experiment.
111 J; refer to page T20.

iii) Calculate the number of moles of ethanol used in the experiment.
0.174 mol; refer to page T20.

iv) Calculate the strength of the hydrogen bond (in kJ mol^{-1}) between ethanol molecules.
1.06 kJ mol^{-1}; refer to page T20.

24 a) Water can be found as ice, water and steam.

The following diagram shows the arrangement of some of the water molecules in a crystal of ice.



With reference to the structure shown above, give ONE reason why ice is less dense than water.
The molecules in ice are held further apart than in liquid water.

b) Water and methane have similar relative molecular masses and both contain the element hydrogen.

The table below gives some information about water and methane.

	H_2O	CH_4
Relative molecular mass	18.0	16.0
Melting point (K)	273	91

i) State the strongest type of intermolecular force holding the water molecules together in the ice crystal. Hydrogen bonds

ii) State the strongest type of intermolecular force in methane. Van der Waals' forces

iii) Give ONE reason why the melting point of ice is higher than the melting point of methane.

(AQA Advanced Subsidiary GCE, Unit 1, Jan. 2011, 1(a)–(b))

Hydrogen bonds are stronger than van der Waals' forces.
More heat is needed to separate the molecules during the melting of ice.

Unit Exercise

25 A student decided to prepare some propene ($CH_3CH{=}CH_2$) from propan-1-ol ($CH_3CH_2CH_2OH$).
Refer to page T20.
Propan-1-ol is a liquid at room temperature whilst propene is a gas. The strongest type of intermolecular forces between propan-1-ol molecules is hydrogen bonding.

a) Draw a diagram to show the hydrogen bonding between TWO propan-1-ol molecules. Include relevant lone pairs and partial charges in your diagram.

b) Propan-1-ol has a higher boiling point than propene. Explain this in terms of intermolecular forces.

(OCR Advanced Subsidiary GCE, Chem. B (Salters), F332, Jan. 2010, 4(a))

26 The boiling points of some hydrides are given below.
Refer to page T21.

Compound	Boiling point (K)
HF	293
HCl	188
HBr	206
HI	238
H_2O	373

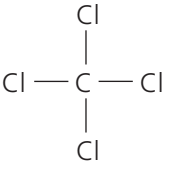
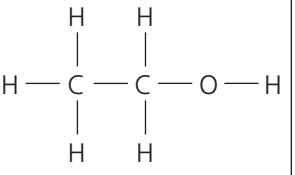
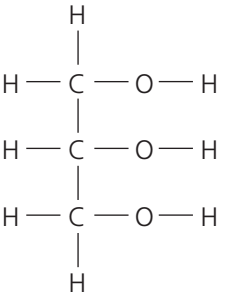
a) Explain, by comparing the forces involved, why HI has a higher boiling point than HBr.

b) Explain, by comparing the types of forces involved, why HF has a higher boiling point than HCl.

c) Suggest why H_2O has a higher boiling point than HF.

(Edexcel Advanced Subsidiary GCE, Unit 2, Jan. 2012, 18)

27 The table below gives the viscosities of three liquids.
Refer to page T21.

Liquid	Structure	Relative viscosity at 20 °C
Tetrachloromethane		*
Ethanol		**
Glycerol		*****

(***** = highest, * = lowest)

Explain why

a) ethanol has a higher viscosity than tetrachloromethane;

b) glycerol has a higher viscosity than ethanol.

3 Electronegativity increases from left to right across a period of the periodic table.
∴ electronegativity of O > electronegativity of N
Within each main group, electronegativity of elements decreases with increasing atomic number.
∴ electronegativity of N > electronegativity of P
The order of increasing electronegativity is: P < N < O.

Topic Exercise

Note: The symbol 🌶 indicates the level of difficulty of a question.

Part I Multiple choice questions

1 Which of the following molecules is linear?

(A) Carbon dioxide (CO_2)
B Sulphur dioxide (SO_2)
C Water (H_2O)
D Methanal (HCHO)

(Edexcel Advanced Subsidiary GCE, Unit 2, Jan. 2010, 2)

2 Which of the following molecules is planar in shape?

(A) BF_3
B NH_3
C CH_4
D PCl_5

(HKDSE, Paper 1A, 2012, 12)

🌶🌶3 Which of the following sequences correctly lists the elements in the order of increasing electronegativity?

Refer to the previous page for explanation.

A N < O < P
B N < P < O
C P < O < N
(D) P < N < O

4 Which of the following molecules is polar?

A BF_3
B C_{60}
(C) NH_3
D SF_6

(HKDSE, Paper 1A, 2012, 5)

5 Which of the following molecules is non-polar?

(A) $BeCl_2$
B NH_3
C H_2O
D HCl

(HKDSE, Practice paper 1A, 2012, 16)

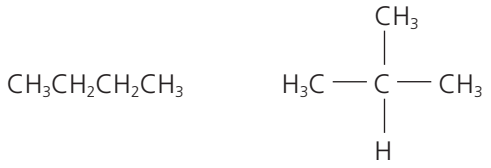
6 Consider the following liquids. Each liquid is run from a burette. Which liquid would NOT be deflected significantly by a charged rod?

A Ethanal (CH_3CHO)
B Ethanol (CH_3CH_2OH)
(C) Tetrachloromethane (CCl_4)
D Trichloromethane ($CHCl_3$)

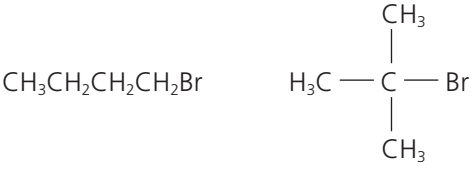
CCl_4 molecules are non-polar. Thus, CCl_4 from a burette would NOT be deflected by a charged rod.

(Edexcel Advanced Subsidiary GCE, Unit 2, Jan. 2011, 7(a))

🌶🌶7 Consider the following compounds, P, Q, R and S.



compound P compound Q



compound R compound S

The boiling points of compounds P, Q, R and S INCREASE in the order

A PQRS.
B RSPQ.
C QSPR.
(D) QPSR.

(Edexcel Advanced Subsidiary GCE, Unit 2, Jun. 2010, 2)

Compared with P and Q, R and S have a greater number of electrons in each molecule. The van der Waals' forces in R and S are stronger than those in P and Q.

Molecule of P has a linear shape while molecule of Q has a more spherical and compact shape. The molecular shape of P allows greater surface contact between molecules. The van der Waals' forces in P are stronger. Thus, the boiling point of P is higher than that of Q.

The boiling point of R is higher than that of S for the same reason.

The boiling points of the compound INCREASE in the order: Q < P < S < R.

8 Which one of the following pairs of liquids, when mixed, would liberate the greatest amount of heat?

A 0.2 mol of trichloromethane ($CHCl_3$) and 0.2 mol of tetrachloromethane (CCl_4)
B 0.2 mol of trichloromethane ($CHCl_3$) and 0.2 mol of propanone (CH_3COCH_3)
C 0.2 mol of tetrachloromethane (CCl_4) and 0.2 mol of butanone ($CH_3COCH_2CH_3$)
D 0.2 mol of propanone (CH_3COCH_3) and 0.2 mol of butanone ($CH_3COCH_2CH_3$)

(HKASLE, Paper 1, 2011, 3(a))

9 To which of the following molecules is / are the 'octet rule' NOT applicable?

(1) OF_2
(2) NO_2
(3) CS_2

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKDSE, Paper 1A, 2013, 23)

10 Which of the following molecules is / are polar?

(1) CO_2
(2) CCl_3Br
(3) PCl_5

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(1) and (3) CO_2 and PCl_5 molecules contain polar bonds. However, they have symmetrical shapes. The bond dipole moments cancel one another out exactly. Thus, these molecules are non-polar.

11 Which of the following statements about NH_3 and PH_3 is / are correct?

(1) Their molecules have a similar shape.
(2) Hydrogen bonding exists in both of them.
(3) The boiling point of PH_3 is higher than that of NH_3.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(2) Hydrogen bonding exists in NH_3 only.
(3) The boiling point of PH_3 is lower than that of NH_3.

Topic Exercise

Directions : Each question (Questions 12–14) consists of two separate statements. Decide whether each of the two statements is true or false; if both are true, then decide whether or not the second statement is a correct explanation of the first statement. Then select one option from A to D according to the following table :

A Both statements are true and the 2nd statement is a correct explanation of the 1st statement.
B Both statements are true but the 2nd statement is NOT a correct explanation of the 1st statement.
C The 1st statement is false but the 2nd statement is true.
D Both statements are false.

	1st statement	2nd statement	
12	Both nitrogen and phosphorus can form pentachloride.	Both nitrogen and phosphorus are Group V elements.	C
13	A silicon tetrachloride molecule has a net dipole moment.	Each Si–Cl bond in a silicon tetrachloride molecule is polar.	C
14	The boiling point of hydrogen chloride is higher than that of hydrogen fluoride.	The molecular size of hydrogen chloride is greater than that of hydrogen fluoride.	C

(HKDSE, Paper 1A, 2013, 24)

12 Nitrogen is in the second period of the periodic table. It cannot form compounds with more than 8 electrons in the outermost shell of its atom. Thus, nitrogen pentachloride does NOT exist.

13 A silicon tetrachloride molecule has a tetrahedral shape. Due to the symmetry of the tetrahedral shape, the four identical Si–Cl bond dipole moments cancel one another out exactly. As a result, the molecule has no net dipole moment.

Part II Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

15 a) For each of the following molecules, draw its three-dimensional structure:

OF_2 and SF_6

b) Suggest why SF_6 exists while OF_6 does not.

(HKASLE, Paper 1, 2011, 2)

16 Both BF_3 and NH_3 exist as simple molecules.

a) For each of these molecules, draw its three-dimensional structure.

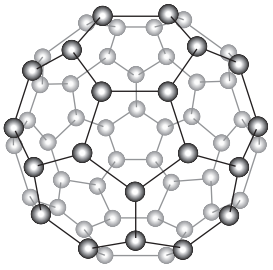
b) For each of these molecules, explain whether or not it is polar.

c) BF_3 reacts with NH_3 to give F_3BNH_3. Describe the bond formation between BF_3 and NH_3.

(HKDSE, Paper 1B, 2013, 2)

17 Buckminsterfullerene is a recently discovered substance.

Look at the diagram below. It shows the structure of buckminsterfullerene.



a) What is the formula for a molecule of buckminsterfullerene? C_{60}

b) Buckminsterfullerene can 'cage' other molecules.
Supply drugs / allow chemical reactions to take place inside the cage.
Describe ONE use of caged molecules.

(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 2, Jun. 2009, 2(a))

18 The arrangement of water molecules in ice is very regular. The shape of individual water molecules is important in building up this regular structure.

a) Draw an electron diagram for a water molecule.

b) Use the electron pair repulsion theory to describe and explain the shape of a water molecule.

(OCR Advanced Subsidiary GCE, Chem. B (Salters), F331, May 2010, 2(e))

19 Explain, from molecular level, why the density of ice is lower than that of water.

(HKDSE, Paper 1B, 2013, 1(c))

20 a) Draw the shape of the NHF_2 molecule and the shape of the BF_3 molecule. Include any lone pairs of electrons that influence the shape.

In each case name the shape.

b) State the strongest type of intermolecular force in a sample of NHF_2.

(AQA Advanced Subsidiary GCE, Unit 1, May 2011, 3(b)(i), (c))

21 Account for each of the following:

a) The boiling point of neon is lower than that of argon.

b) $NH_3(g)$ is very soluble in water, but $PH_3(g)$ is almost insoluble.

(HKASLE, Paper 2, 2010, 5(a)–(b))

22 Explain, in terms of intermolecular forces, why chlorine is a gas at room temperature and pressure but bromine is a liquid under the same conditions. As part of your answer, you should explain how the intermolecular forces arise. Refer to page T22.

(OCR Advanced Subsidiary GCE, Chem. B (Salters), F332, Jun. 2009, 1(e))

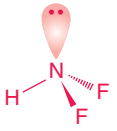
18 a) 

b) A water molecule has two lone pairs and two bond pairs of electrons in the outermost shell of the oxygen atom.

The electron pairs repel one another and stay as far apart as possible. The four pairs of electrons in the molecule will adopt a tetrahedral arrangement.

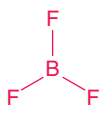
The shape of a molecule is determined only by the arrangement of atoms. Thus, the water molecule is V-shape or bent shape.

20 a) NHF_2 molecule



trigonal pyramidal shape

BF_3 molecule



trigonal planar shape

b) Hydrogen bonds

23 The table below shows the boiling points of some hydrogen compounds formed by Group VI elements.

Refer to page T22.

	H_2O	H_2S	H_2Se	H_2Te
Boiling point (K)	373	212	232	271

a) State the strongest type of intermolecular force in water and in hydrogen sulphide (H_2S).

b) Draw a diagram to show how TWO molecules of water are attracted to each other by the type of intermolecular force you stated in part (a). Include partial charges and all lone pairs of electrons.

c) Explain why the boiling point of water is much higher than the boiling point of hydrogen sulphide.

d) Explain why the boiling points increase from H_2S to H_2Te.

(AQA Advanced Subsidiary GCE, Unit 1, Jan. 2011, 3)

24 Solids exist as lattice structures.

The solid lattice structure of ammonia, NH_3, contains hydrogen bonds.

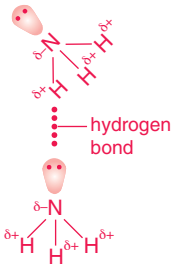
a) Draw a diagram to show hydrogen bonding between TWO molecules of NH_3 in a solid lattice.

Include relevant dipoles and lone pairs.

b) Suggest why ice has a higher melting point than solid ammonia.

(OCR Advanced Subsidiary GCE, Chem. A, F321, Jun. 2011, 5(b))

24 a)



b) A hydrogen bond requires both a lone pair of electrons on an atom of N, O or F, and a hydrogen atom.

In ice, there are two hydrogen atoms in each molecule and two lone pairs of electrons on the oxygen atom, allowing the formation of two hydrogen bonds.

In solid ammonia, there are three hydrogen atoms in each molecule, but there is only one lone pair of electrons on the nitrogen atom. This means that each molecule can only form one hydrogen bond.

Thus, more heat is required to break the hydrogen bonds in ice during melting.

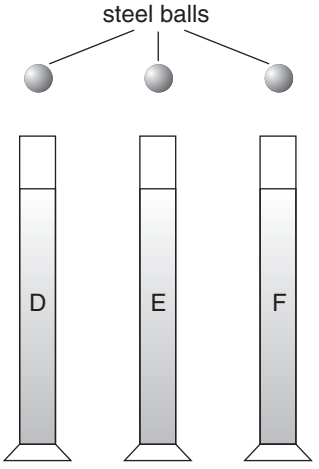
25 The table below lists some information about six hydroxyl compounds, A, B, C, D, E and F:

Compound	Structural formula	Boiling point at 1 atm (°C)	Density at 20 °C (g cm^{-3})
A	CH_3OH	65	0.7914
B	CH_3CH_2OH	78	0.7893
C	$CH_3CH_2CH_2OH$	97	0.8035
D	$CH_3CH(OH)CH_3$	82	0.7855
E	$HOCH_2CH_2CH_2OH$	213	1.0597
F	$HOCH_2CH(OH)CH_2OH$	290	1.2613

a) Account for the variation in boiling points of compounds A, B and C.

b) Explain why the density of compound C is greater than that of compound D.

c) Three identical steel balls are added separately to three identical vertical glass tubes each containing the same volume of compounds D, E and F as shown in the diagram below.

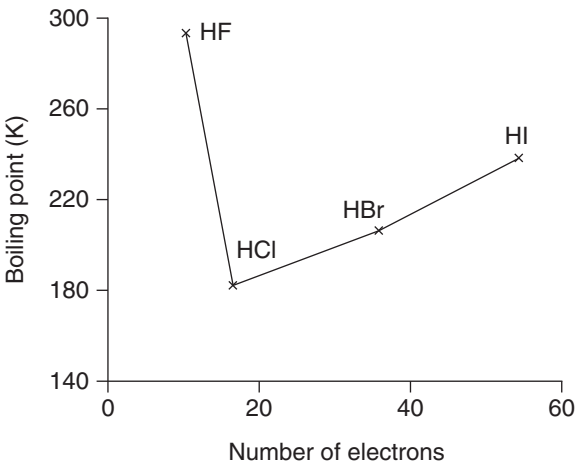


In which tube will the steel ball take the longest time to reach the bottom? Explain your answer. (You are required to consider the intermolecular attraction forces involved.)

(Communication mark will be awarded for this part.)

(HKDSE, Practice paper 1B, 2012, 6(b)–(d))

26 The graph below shows the boiling points of the hydrides of Group VII.



a) i) Identify the type of intermolecular force that gives rise to the unusually high boiling point of hydrogen fluoride.

Hydrogen bond

ii) State whether the electronegativity of fluorine is greater than, similar to or less than, that of bromine.

Fluorine is more electronegative.
So, the H–F bond is more polar.

Hence explain why hydrogen fluoride can form the type of intermolecular force named in (a)(i) but hydrogen bromide cannot.

iii) Use the graph to predict what the boiling point of hydrogen fluoride would be without the presence of the type of intermolecular force named in (a)(i).

Between 150–180 K

b) Propanone, CH_3COCH_3, is a useful solvent for cleaning glassware in laboratories.

Propanone can be used to remove both water and octane (C_8H_{18}) from glassware.

For each of these substances, identify the strongest intermolecular force formed with propanone and the feature of the propanone molecule involved.

(Edexcel Advanced Subsidiary GCE, Unit 2, Jun. 2009, 21)

Water: Hydrogen bonds formed between the hydrogen atoms of water molecules and the oxygen atoms of propanone.

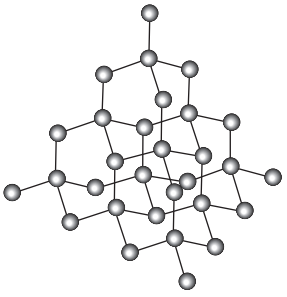
Octane: Van der Waals' forces exist between the CH_3 groups of propanone and the CH groups of octane.

27 Both ethanol and water contain hydrogen bonds. By considering the hydrogen bonding in these two solvents, suggest why 2-chlorobutane ($CH_3CH_2CHClCH_3$) is more soluble in ethanol than in water.

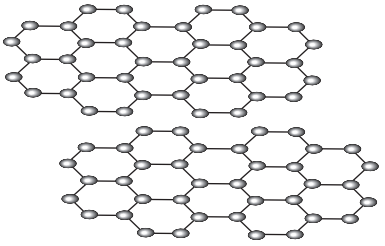
(Edexcel Advanced Subsidiary GCE, Unit 2, Jan. 2010, 18(f)(ii))

28 The diagrams show the structures of diamond and graphite. They are different structural forms of the element carbon.

Refer to page T23.



diamond

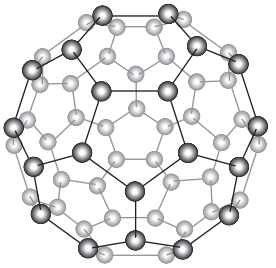


graphite

a) What type of structure are both diamond and graphite?

b) Graphite can be used as a lubricant. Explain why.

c) During the twentieth century, another structural form of carbon was discovered. In this structural form, the molecules have the formula C_{60} and are shaped like footballs.



i) C_{60} has a much lower melting point than diamond. Explain why.

ii) Would you expect C_{60} to act as a lubricant? Explain your answer.

(Edexcel IGCSE, Chemistry, Paper 2, Nov. 2008, 10(a), (c)–(d))

27 Water molecules can form an average of 2 hydrogen bonds per molecule while ethanol molecules can form an average of 1 hydrogen bond per molecule only.

More heat is needed to break the hydrogen bonds in water. / Hydrogen bonding in water is stronger.

So, 2-chlorobutane is less soluble in water.